

Project Management

ENGR. JUVEN G. LUZORATA



PERT/CPM

- a project management tool in the construction industry to **plan**, **schedule**, and **control** projects
- a project network





Project Network



consists of **node** and **arc** forming a network used to represent projects

Project's Basic Information

Activity Information

Detail of the activity (e.g., excavation, masonry, finishing)

Precedence Relationship

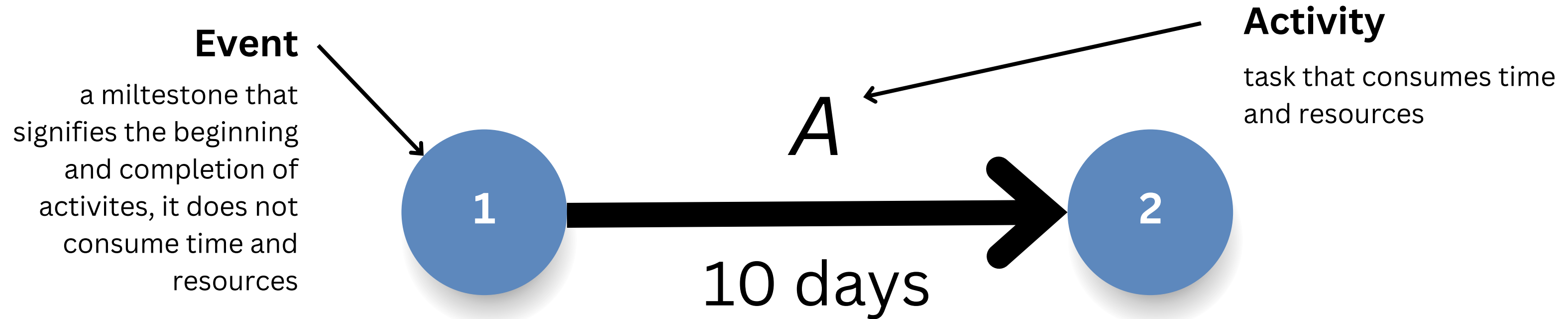
Identification of immediate predecessor

Time information

Estimate of the activity duration

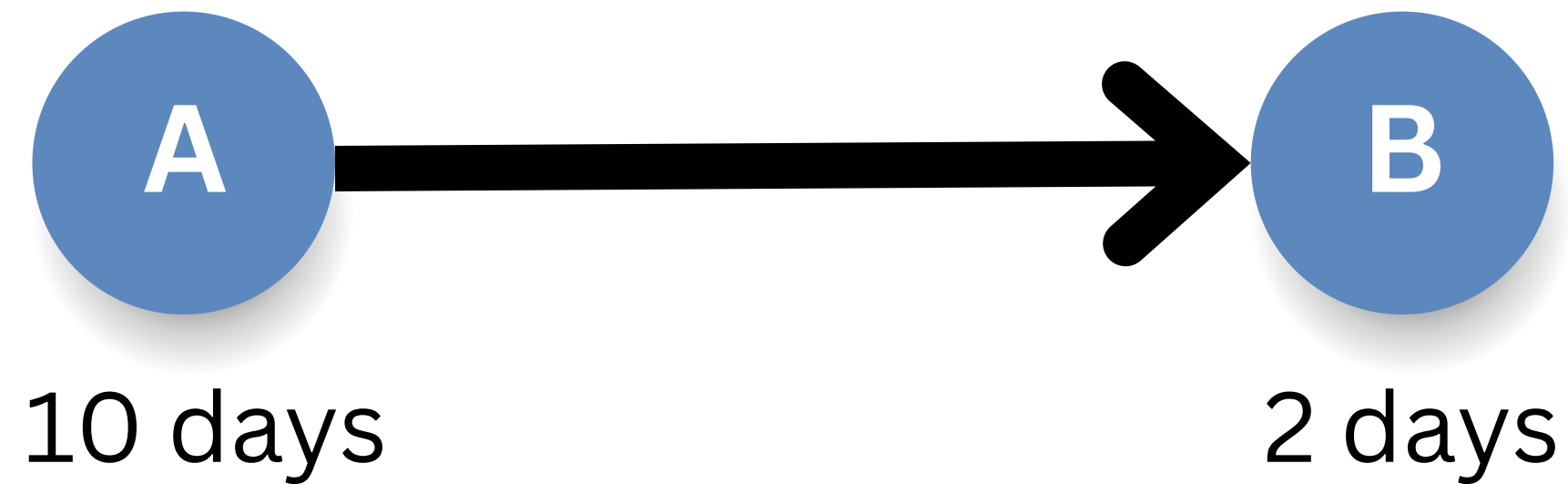
Classification of Network

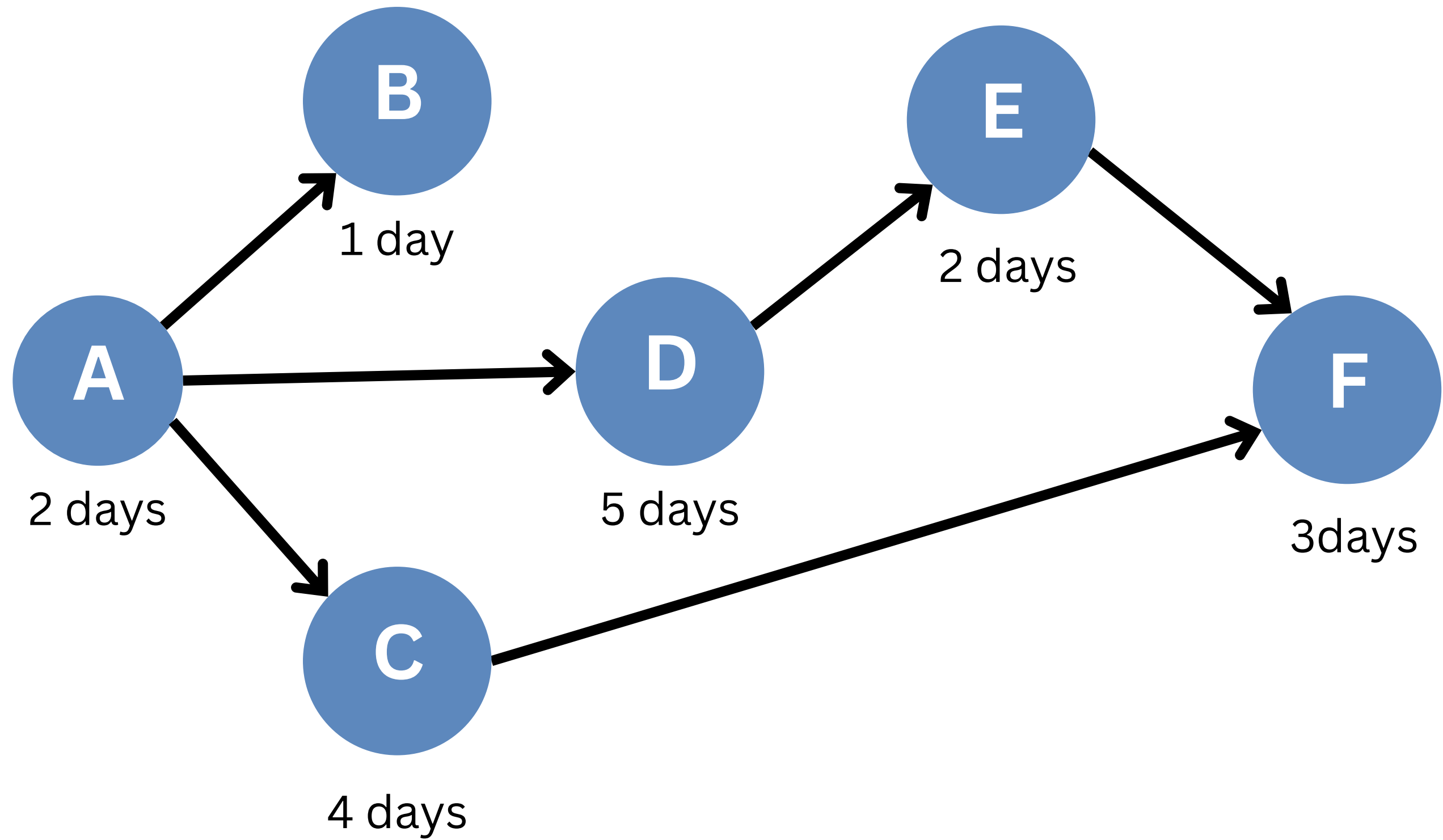
Activity-on-Arc

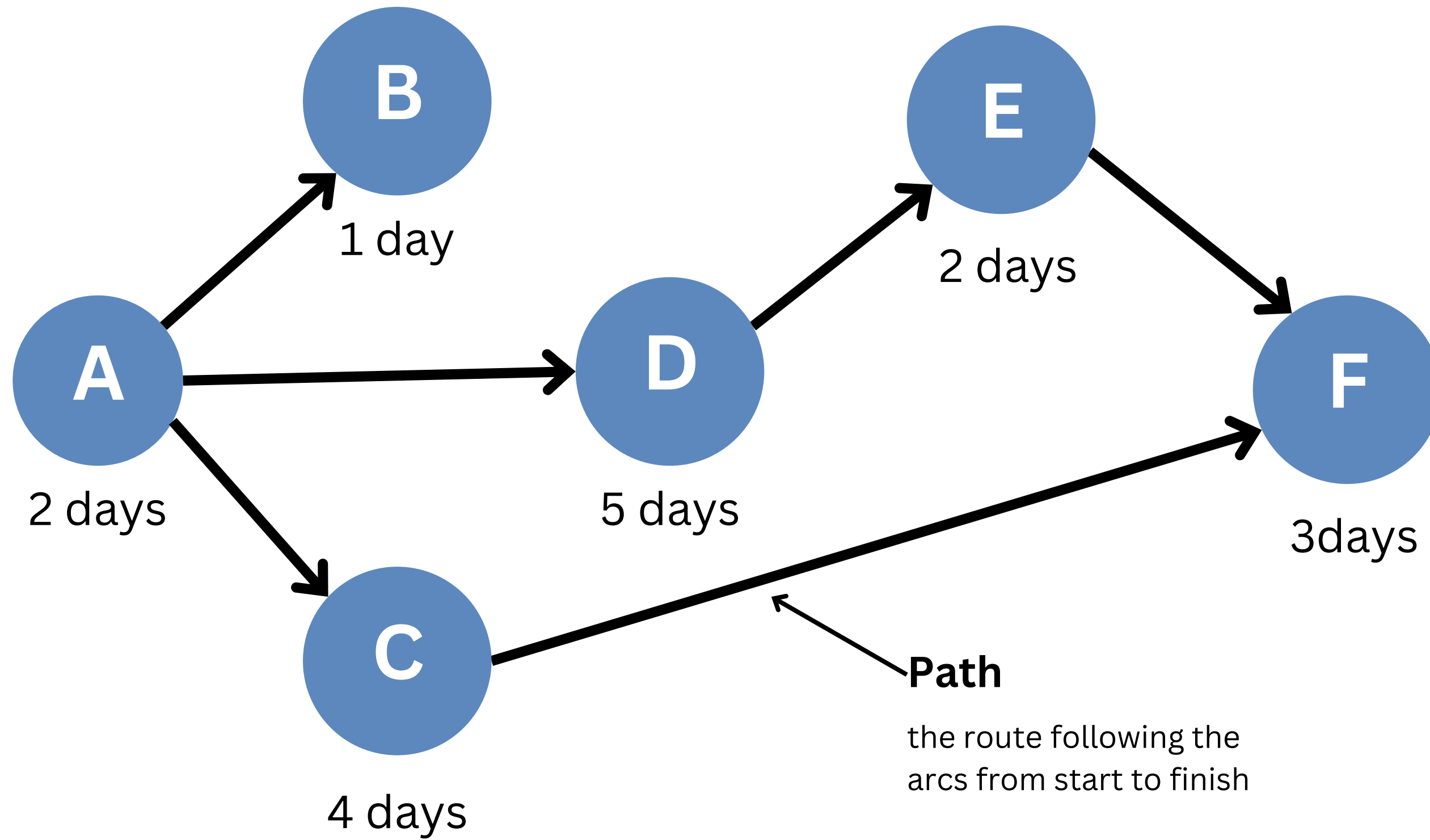


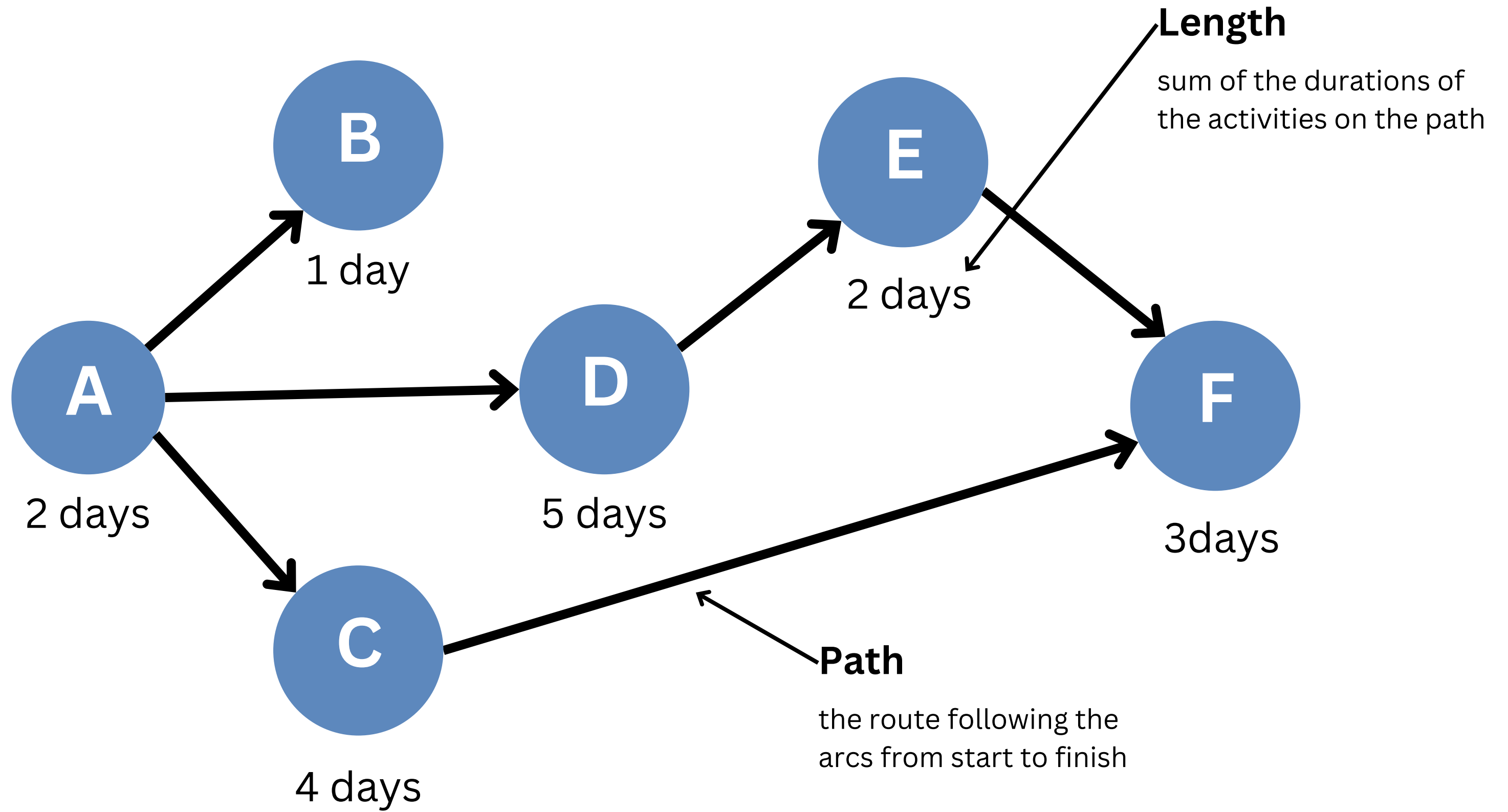
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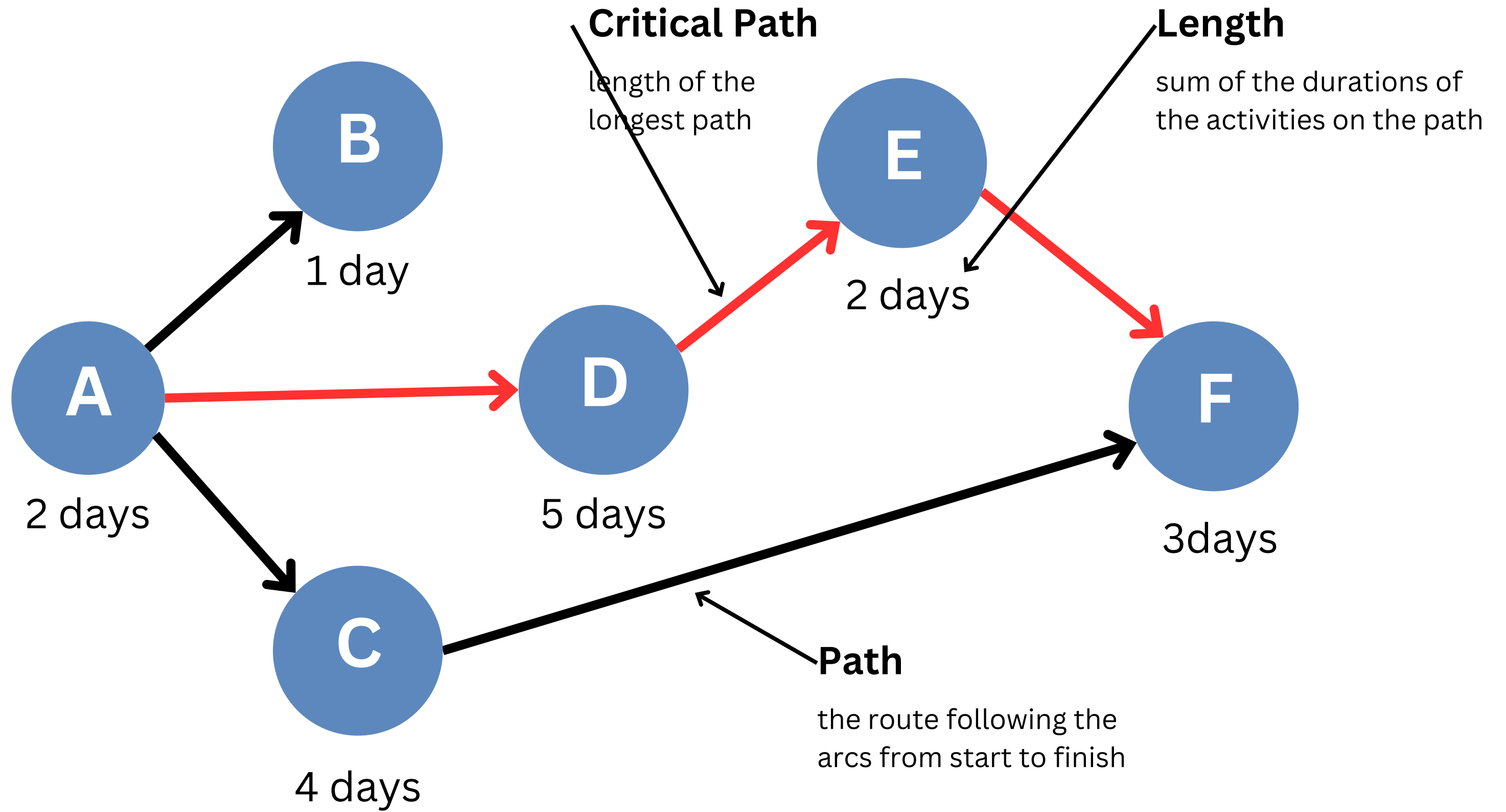
Activity-on-Node

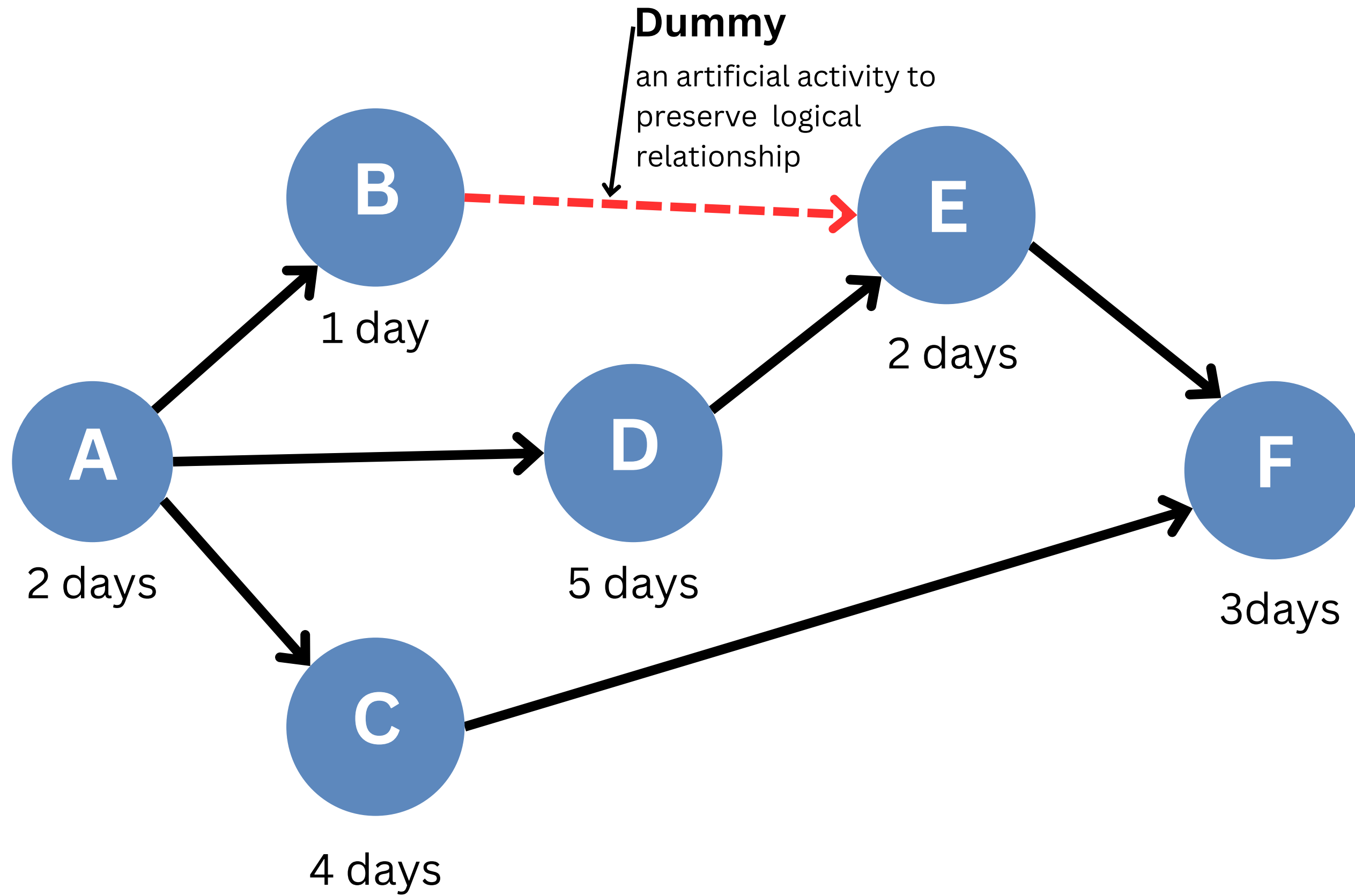


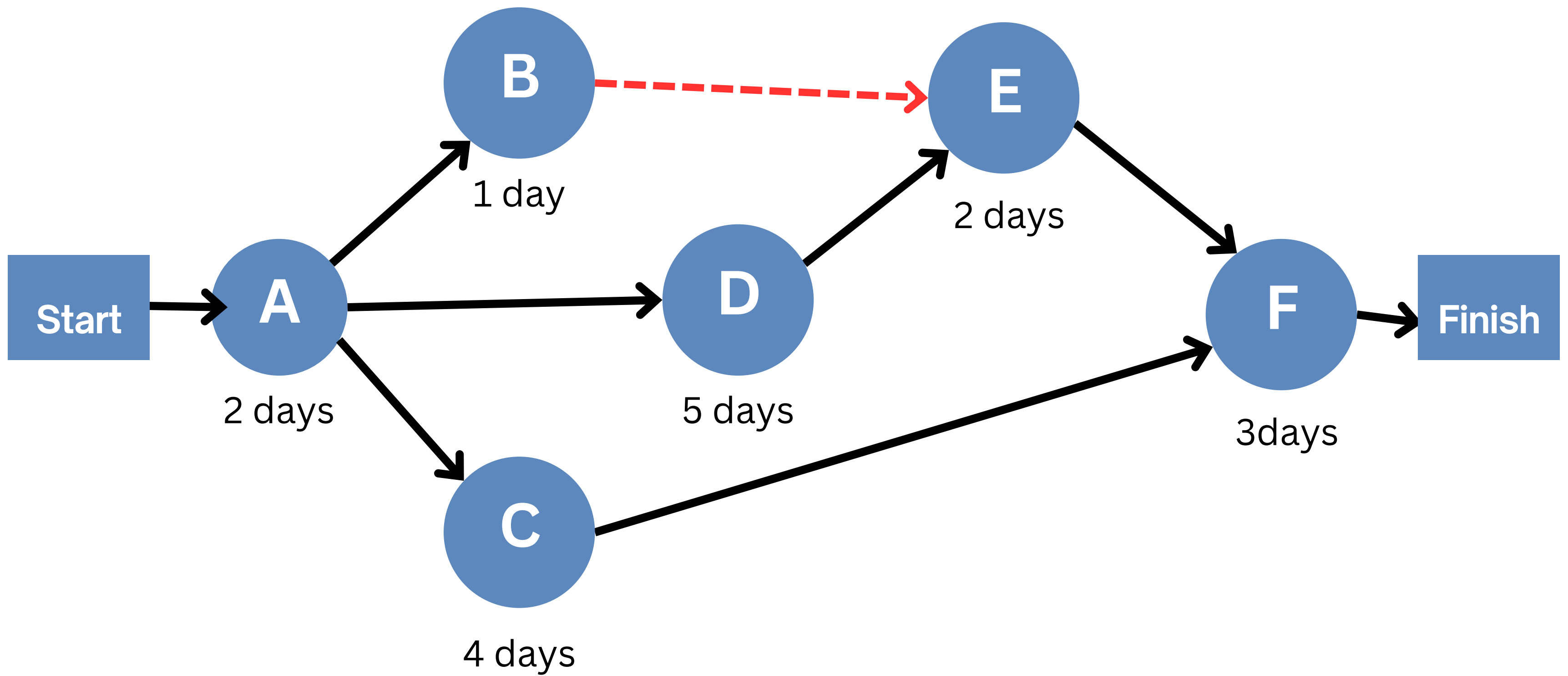












Forward Pass

assumes that the estimated activity duration is followed

**Early Start
Early Finish**

earliest possible start and finish time for a project without delay

Backward Pass

determines how much later than indicated an activity start or finish without delaying the project

**Late Start
Late Finish**

latest possible start and finish time for a project without delay

Slack or Float

allowable time for an activity to delay without delaying the overall project duration

Critical Path

has zero slack or float, delaying an activity in the critical path will delay the overall project duration



PERT

Program Evaluation and Review Technique

- probabilistic approach
 - it is **event oriented**
 - it considers time uncertainty for completion of an activity
- | | |
|-----------------------------|---|
| Most Likely Estimate | most likely value |
| Optimistic Estimate | duration under favorable conditions |
| Pessimistic Estimate | duration under not favorable conditions |
- What is the probability of finishing a project at a certain duration?

CPM

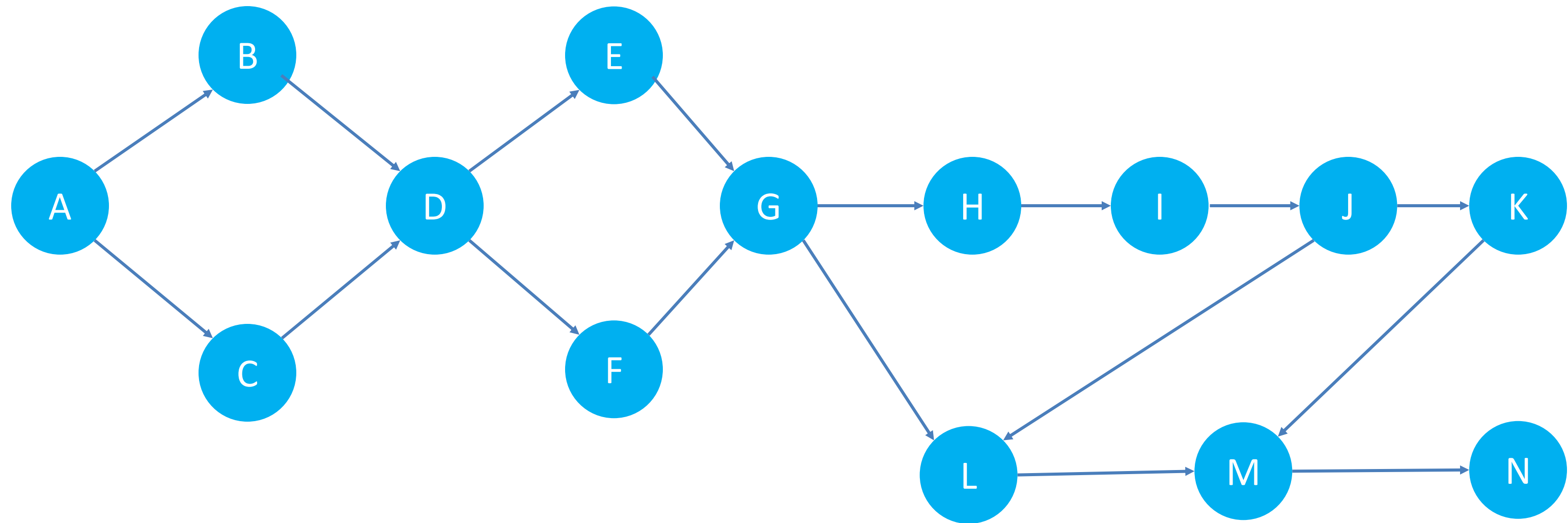
Critical Path Method

- deterministic approach
 - it is **activity oriented**
 - it assumes that activity follows estimated activity duration
- | |
|---------------------------|
| Early Start/Finish |
| Late Start/Finish |
- **crashing** - reducing the duration at a cost
 - What is the time-cost tradeoff?



St. Gerard Construction Company was awarded by the DPWH a flood control project contract amounting to 100 billion.

A	Mobilization	-	2 weeks
B	Survey & Staking	A	1 week
C	Environmental Control	A	1 week
D	Diversion construction	B, C	2 weeks
E	Dewatering & Flow bypass	D	1 week
F	Channel clearing	D	2 weeks
G	Excavation	E, F	3 weeks
H	Subgrade preparation	G	1 week
I	Formwork & rebar	H	2 weeks
J	Concrete lining	I	3 weeks
K	Curing & Surface finishing	J	2 weeks
L	Slope protection	G, J	3 weeks
M	Backfilling & compaction	K, L	2 weeks
N	Final grading, cleanup, & turnover	M	1 week



How will you rate their works?





CPES

Constructor's Performance Evaluation System

- a system of grading the performance of a constructor for a specific kind of projects using a **set of criteria**

Constructor

a builder, having a capacity to undertake or submit a bid for infrastructure projects

CPE

a Constructor's Performance Evaluator



CPE Minimum Qualification

- a licensed **Engineer** or **Architect**
- at least **5 years of experience** in construction industry
- not convicted of a crime for **moral turpitude** or a crime with a penalty of **6 months imprisonment**
- willing to undergo all screening requirements and accreditation course for CPEs





Corrective Action Request

a request form where the CPE records his/her validated findings of the constructor's noncompliance with any checklist and requiring immediate action

Corrective Action Plan

proposed action plan to correct non-conformance findings including specific action and timetable to immediately correct findings

Spot

area selected from identified location of on-going and completed items of work which will be subject for evaluation/inspection



During Construction

60%

**Horizontal
Projects**

- Road and Bridges
- Ports and Harbors
- Irrigation and Flood Control
- Water Supply and Sewerage
- Mooring Facilities for Power Barges

70%

**Vertical
Projects**

- Housing
- Building
- Power Transmission Line
- Substation and Diesel Power Plant

Upon Completion

40%

**Horizontal
Projects**

- Road and Bridges
- Ports and Harbors
- Irrigation and Flood Control
- Water Supply and Sewerage
- Mooring Facilities for Power Barges

30%

**Vertical
Projects**

- Housing
- Building
- Power Transmission Line
- Substation and Diesel Power Plant



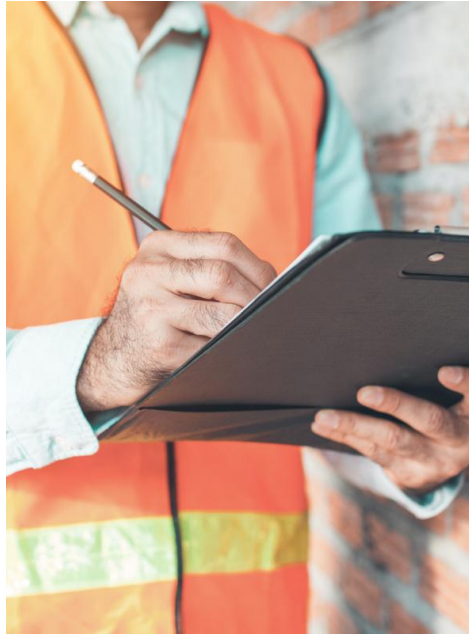
Number of Visits

During

- **Minimum 2 visits**, when the actual accomplishment is at least **30%**
- **1 visit** for projects having 90 calendar days and below

Upon Completion

- **final visit** for 100% completion



CPES Criteria

During

Upon
Completion

Workmanship

0.40

0.50

Materials

0.30

0.20

Time

0.15

0.30

Facilities

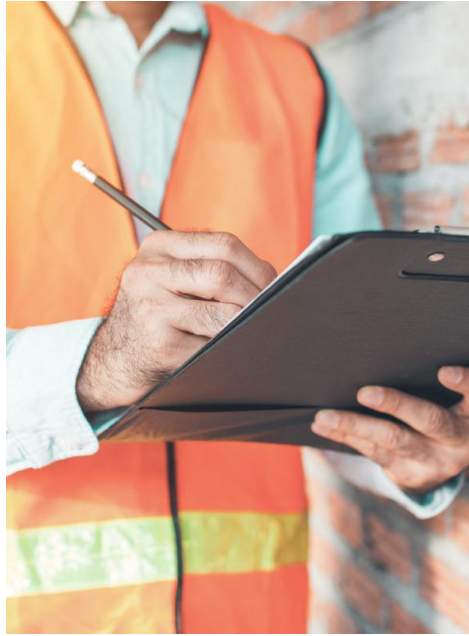
0.03

Environment, Safety & Health

0.07

Resource Deployment

0.05



Scoring Remarks

Adjective Rating

Outstanding
Very Satisfactory
Satisfactory
Unsatisfactory
Poor

Rating

> 96%
> 89 - 96%
> 82 - 89%
75 - 82%
< 75%

CPES in Construction Industry

- CPES rating shall form part for PCAB registration
- a Constructor shall have at least satisfactory rating for agency shortlisting
- a Constructor shall have at least satisfactory rating and no zero rating for Time in agency's post qualification
- a Constructor's final rating of Poor or Unsatisfactory is blacklisted



PCAB License Category

Philippine Construction Authority Board

Size Range	Size Range	Single Large Project	Allowable Range of Contract Cost
Large B	AAAA and AAA	> 250 Million	less or above 450 Million
Large A	AA	150 - 250 Million	Up to 450 Million
Medium B	A	75 - 150 Million	Up to 300 Million
Medium A	B	15 - 75 Million	Up to 150 Million
Small B	C and D	<= 15 Million	Up to 30 Million
Small A	E	Up to 1 Million	Up to 11 Million



Forest Engineering Environment Engineering

ENGR. JUVEN G. LUZORATA

Forest Engineering

- an engineering approach or process of managing timber extraction and afforestation or reforestation practices

Timber Extraction

Felling

cutting down
trees

**Log
Processing**

delimbing,
debarking,
slashing

Skidding

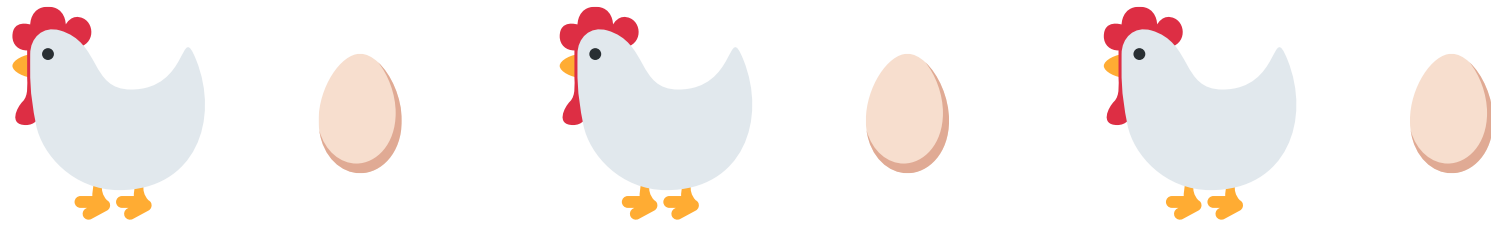
from cutting point
to the road

Transportation

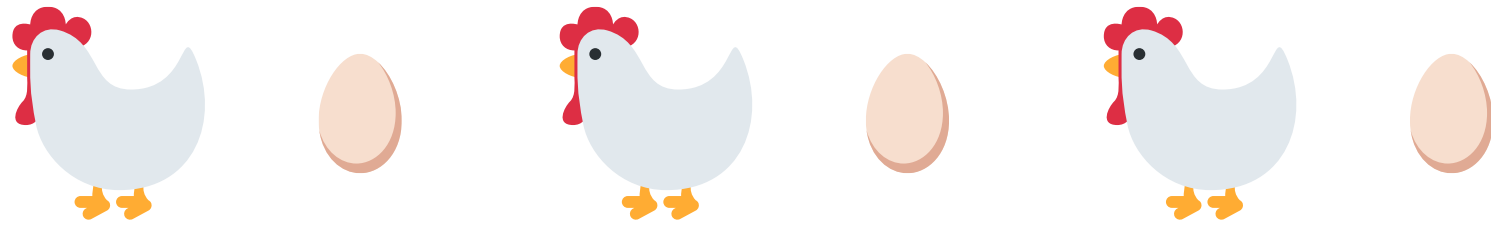
from the road to
sawmill

A man with dark hair, wearing a white button-down shirt, is pointing his right index finger upwards. He has a serious expression and is looking slightly to the right of the camera. The background is plain white.

**neeeever
mag-forget ng
basic termabols**

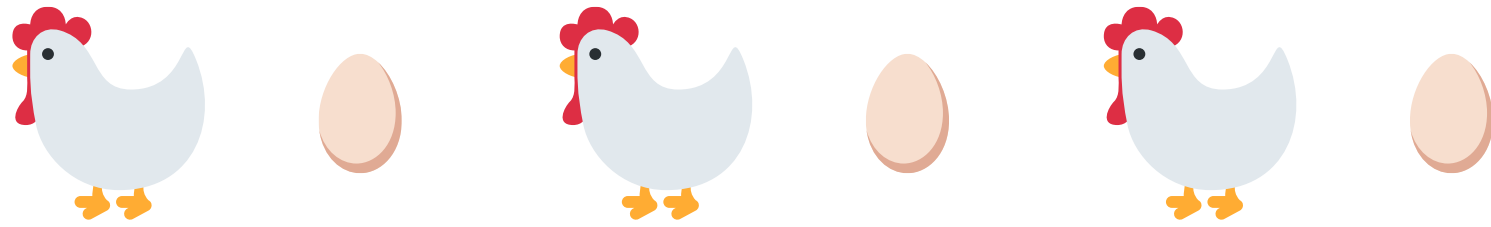


Planting trees in areas that have never
been forested before

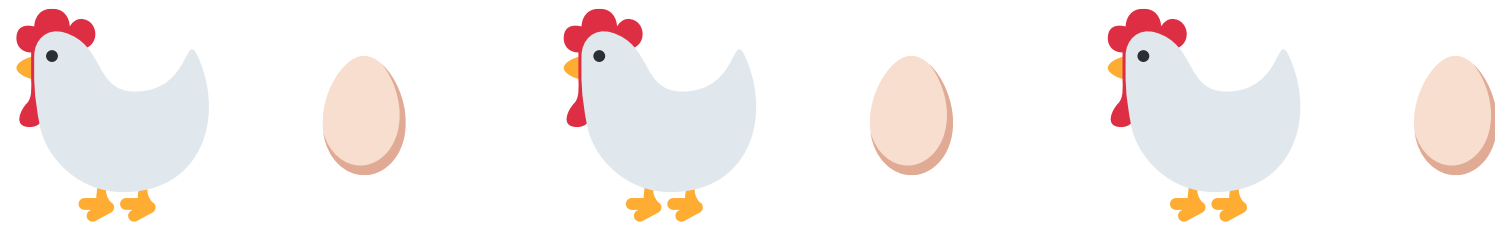


Planting trees in areas that have never
been forested before

Afforestation



Timber harvesting process.



Timber harvesting process.

**Felling - Log Processing - Skidding -
Transporting**

**pov: nagkatitigan ang mga
pumaldo sa terms**



Clearclutting

a logging practice where all or most of the trees in a specific area are cut down at once



Thinning

a selective cutting practice where only some trees are removed, usually the weaker, overcrowded, or less desirable ones



Softwood

a gymnosperm (cone-bearing trees)
with simple cell structures (e.g., pine,
pruce, cedar)



Hardwood

an angiosperm (broadleaf trees) with
vessels that make their wood
structure more complex



Wood

- a **lignocellulosic material**, composed of cellulose, hemicellulose, and lignin
- **lignin** binds the cellulose to make it stronger
- a **hygroscopic material**



Lumber

processed log or wood

Slab

rough lumber cut tangent to annual
ring



Dressed Lumber

planed lumber having at least one smooth side

- **S2S** - smooth on two sides
- **S4S** - smooth on four sides



Timber

a piece of lumber 5" or larger in its smallest dimension



Plank

wide piece of lumber from **4 - 5"** thick



Board

a piece of lumber less than **4 cm thick** and at least **10 cm wide**



Flitch

thick piece of lumber or thick slab that is unprocessed



Flitch

thick piece of lumber or thick slab that is unprocessed



Wood Defects

➤ Heart shakes

radial crack originating from the heart of the tree

➤ Star shakes

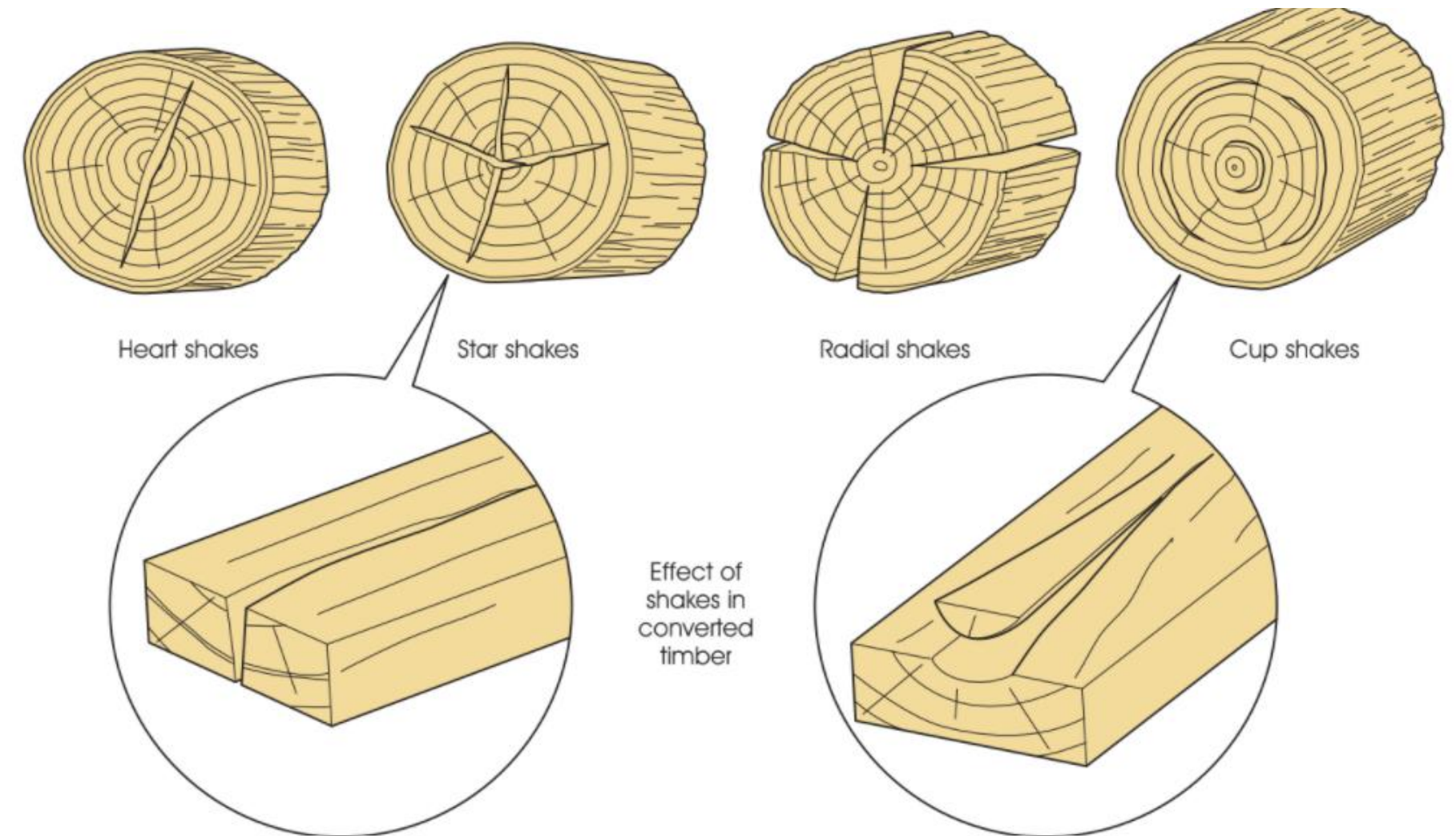
several heart shakes

➤ Wind shake or Cup shake

cracks or breaks across annual rings of the wood during its growth caused by wind pressure

➤ Knot

occurs at the starting point of a limb or branch



Wood Products

➤ Plywood

comprising thin layers of wood (veneers or piles)

➤ Particle Board

made from chips, sawdust, and shavings, and is low density

➤ Fiberboard

made from fine wood fibers, and is medium to high density

➤ Glulam

Engineered wood made by gluing layers of lumber to create stronger and larger pieces than solid wood.



Yard lumber

- for light construction and finishing work

Shop lumber

- for doors, cabinets, and furniture

Structural lumber

- lumber used for beams and columns

Profiles

➤ flooring and cladding profiles

cladding means a nonstructural covering material applied to the exterior of a building to provide protection and aesthetic finish



T&G



V-Rustic

a. T&G

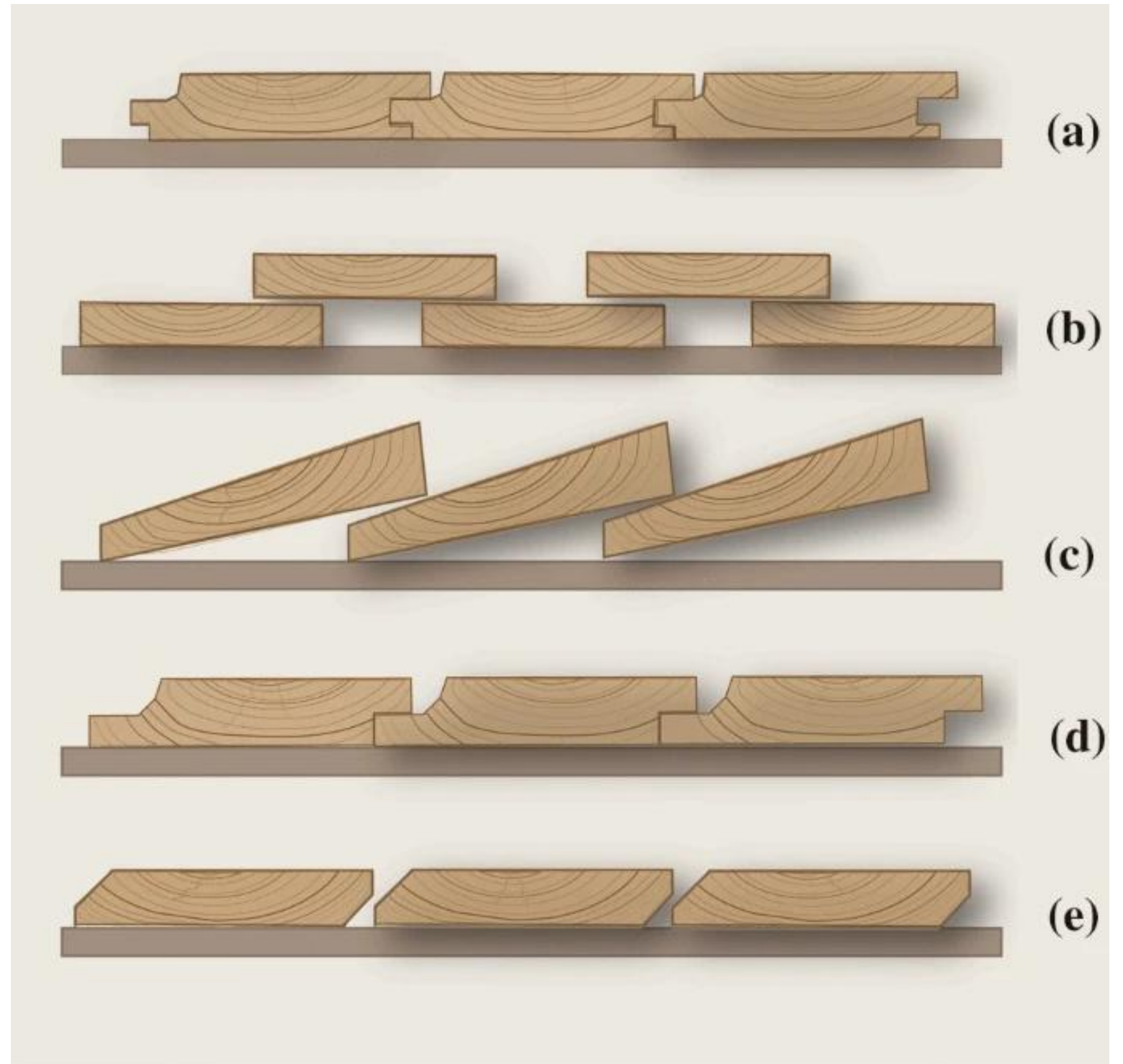
b. rectangular

c. featheredge

d. shiplap

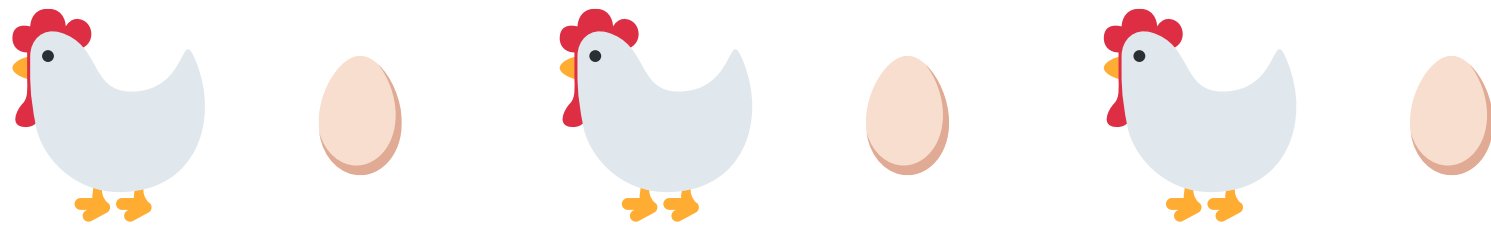
e. parallelogram

f. v-rustic

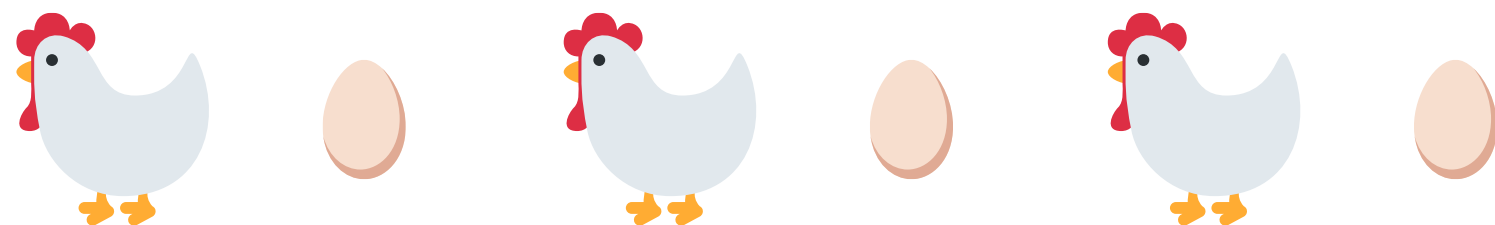


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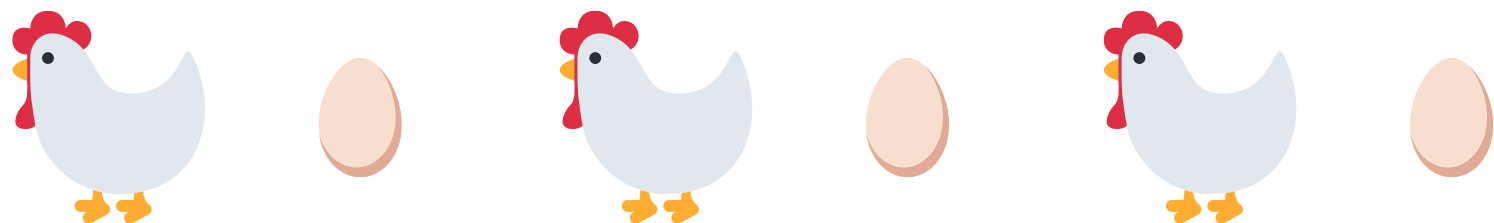


Selective cutting practice

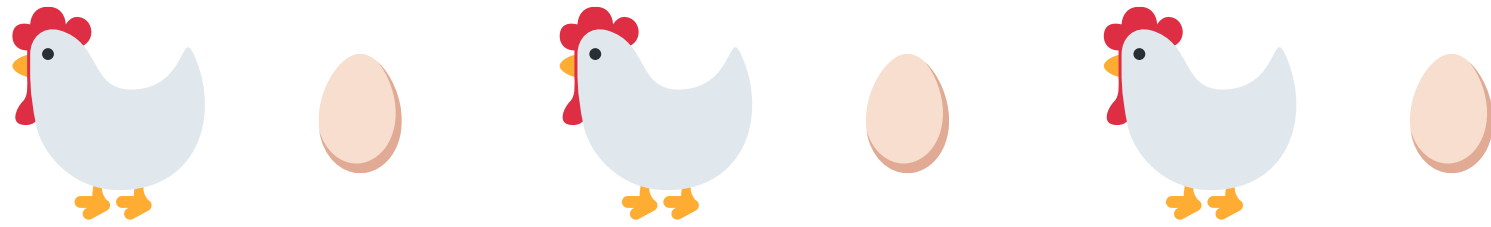


Selective cutting practice

Thinning

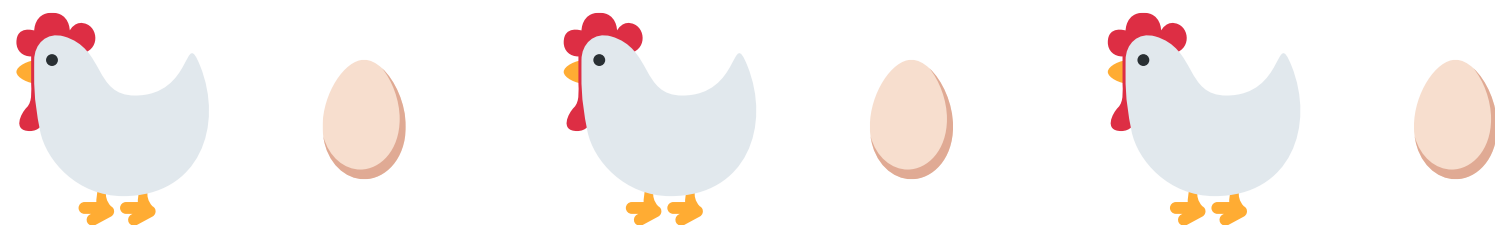


All trees are cut down

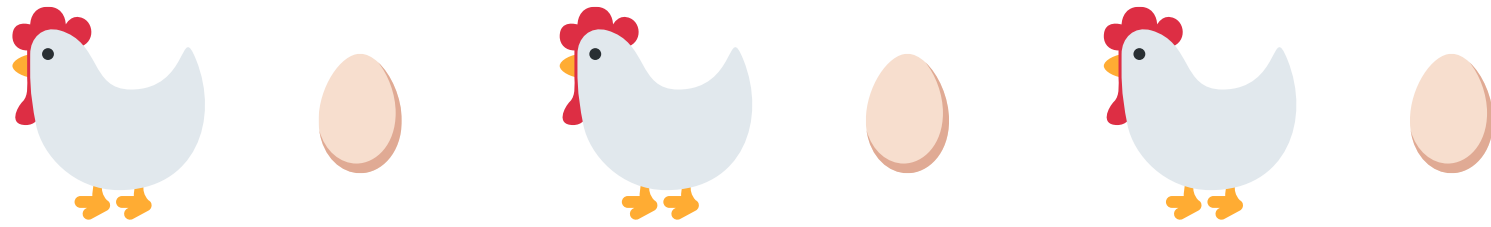


All trees are cut down

Clearcutting

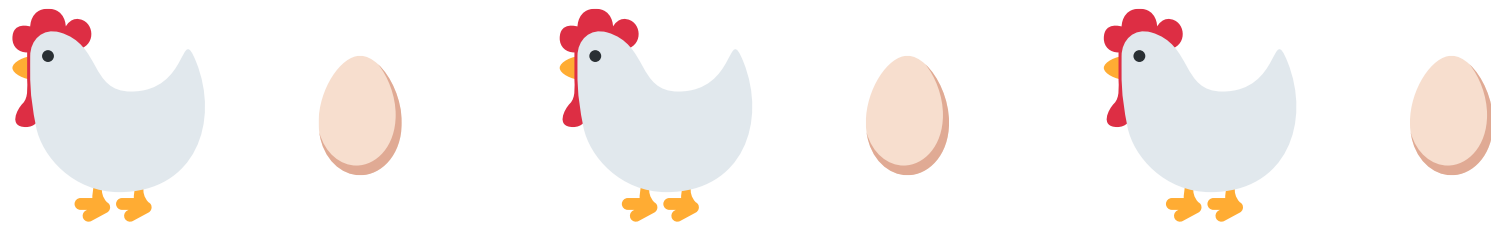


A thick piece of lumber

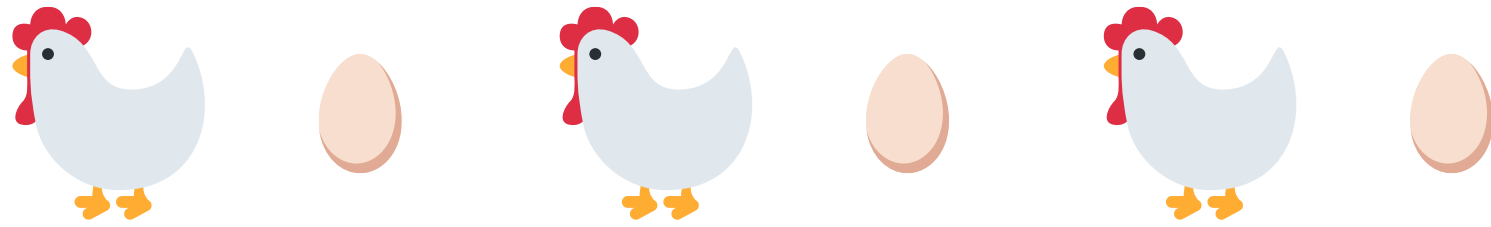


A thick piece of lumber

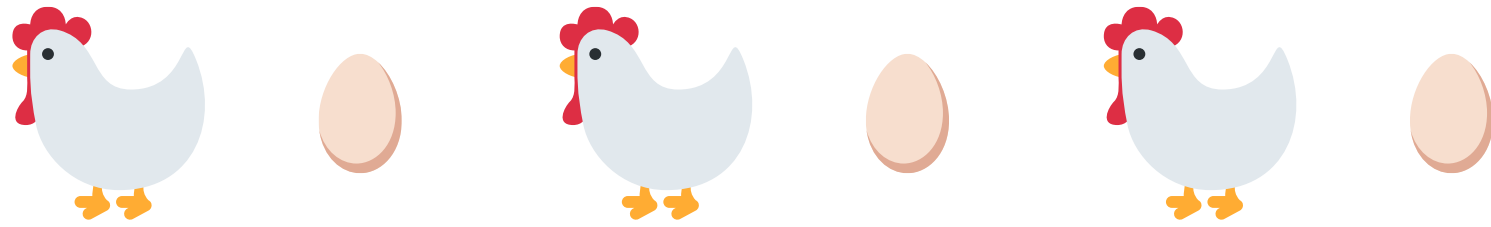
Flitch



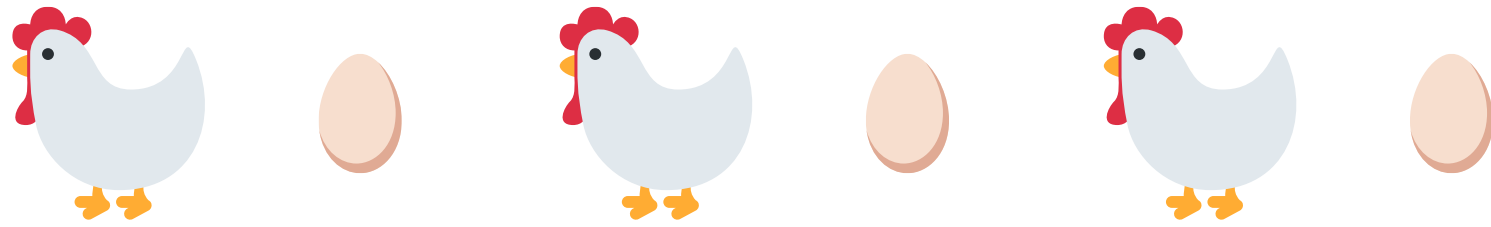
An angiosperm tree



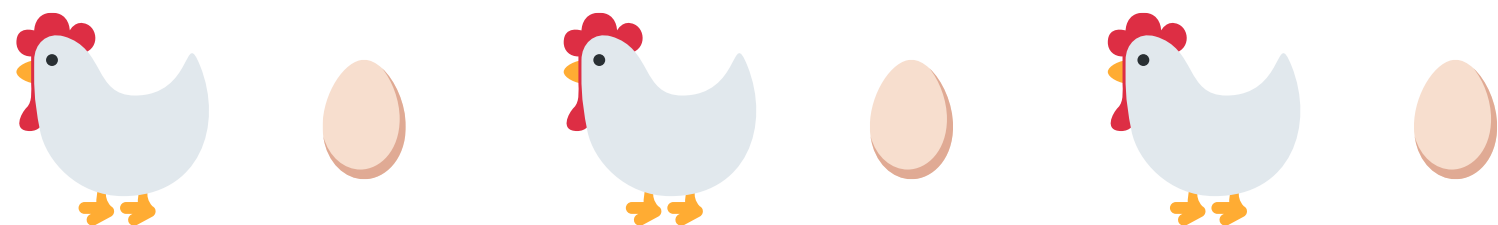
An angiosperm tree
hardwood



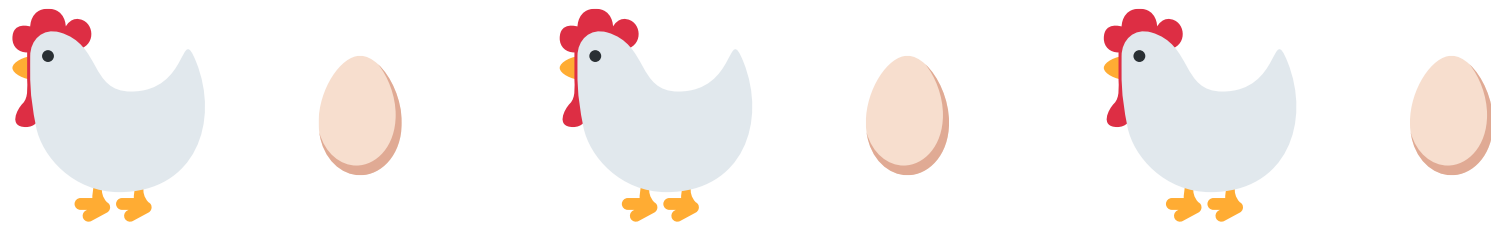
A gymnosperm



A gymnosperm
softwood

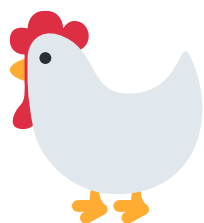
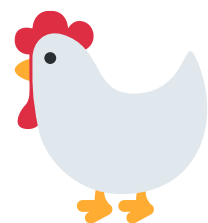
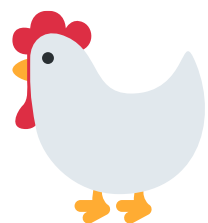


breaks or cracks across annual rings

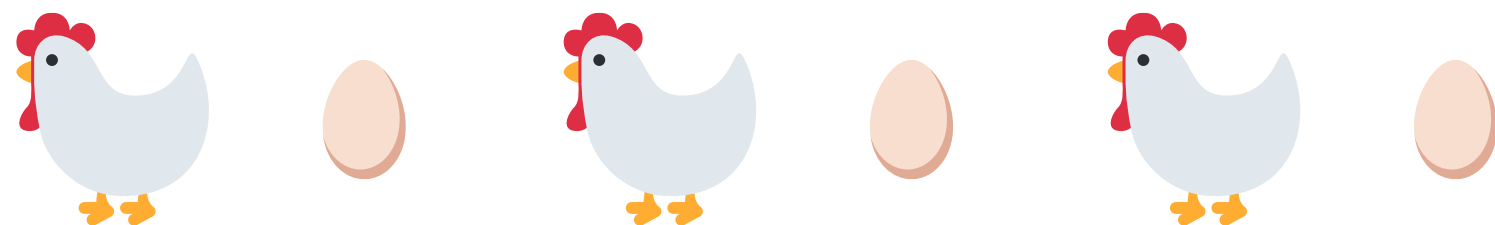


breaks or cracks across annual rings

wind shake or cup shake

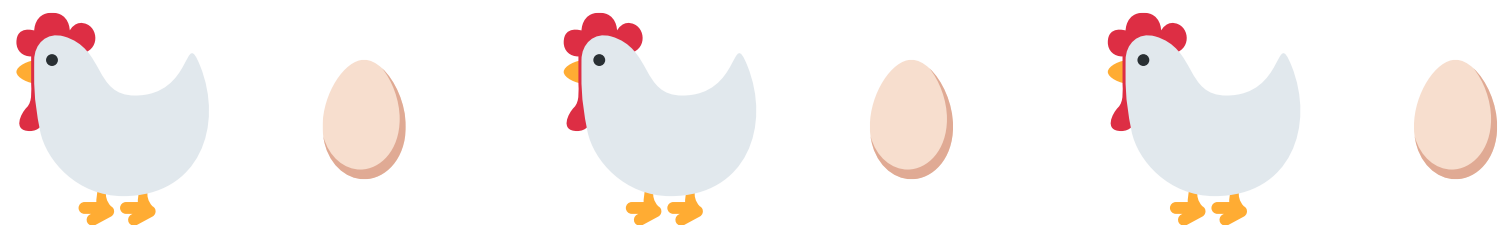


identify

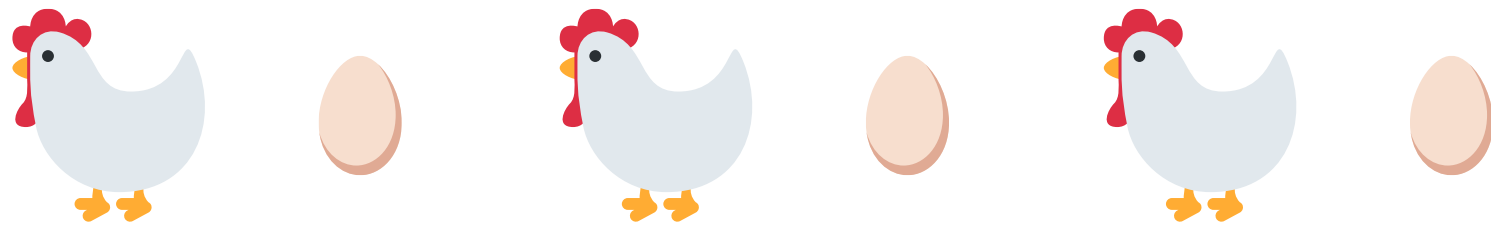


identify

wind shake or cup shake

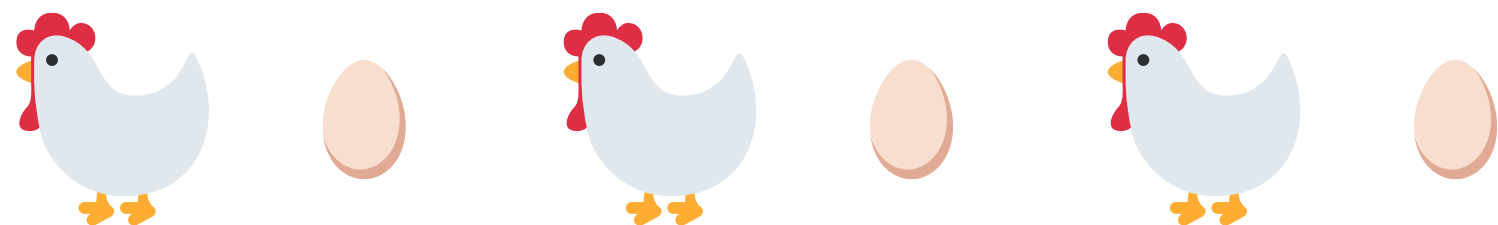


wood binding agent

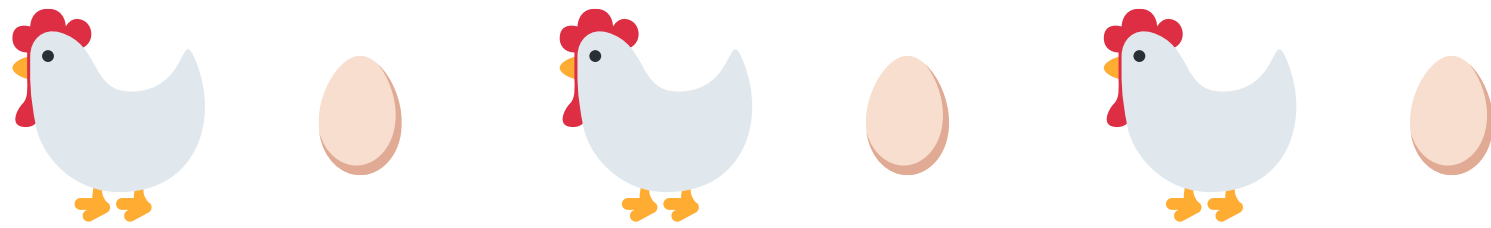


wood binding agent

lignin

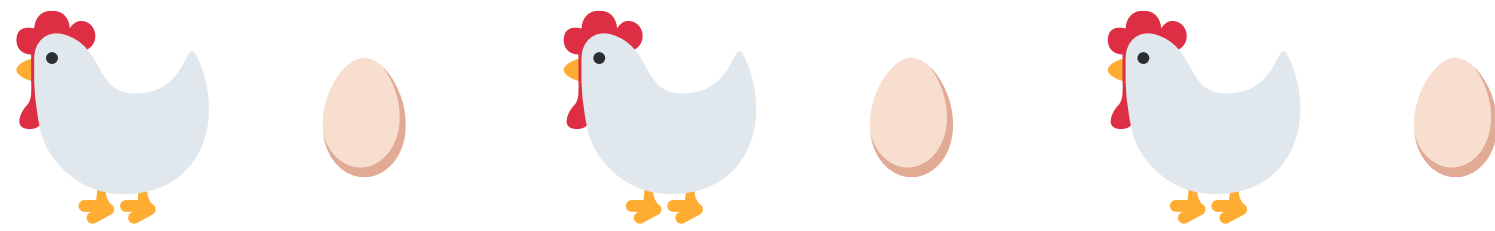


wood binding agent

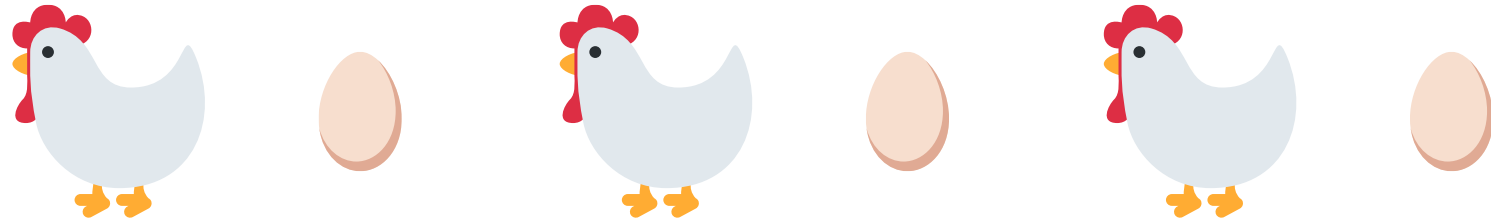


wood binding agent

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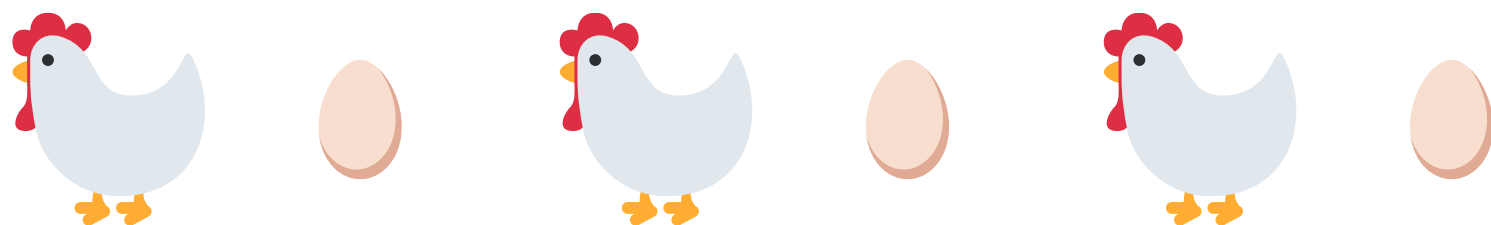


primary component of wood cell

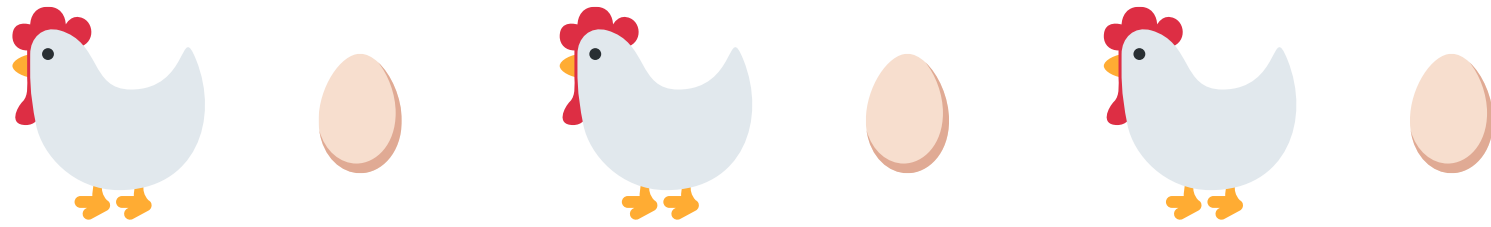


primary component of wood cell

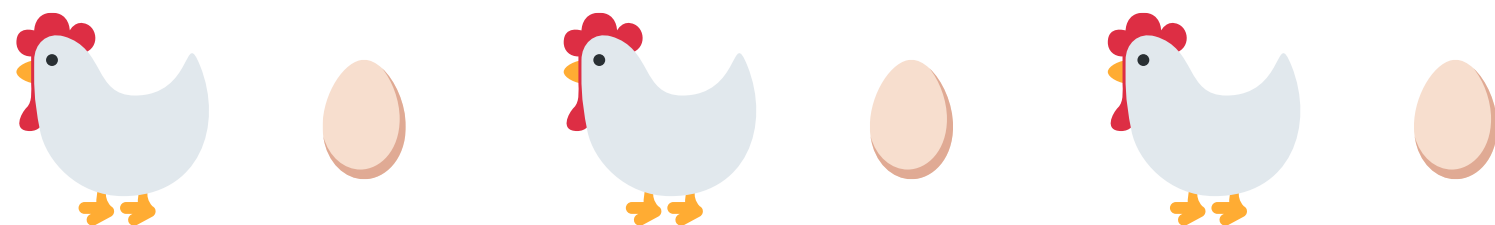
cellulose



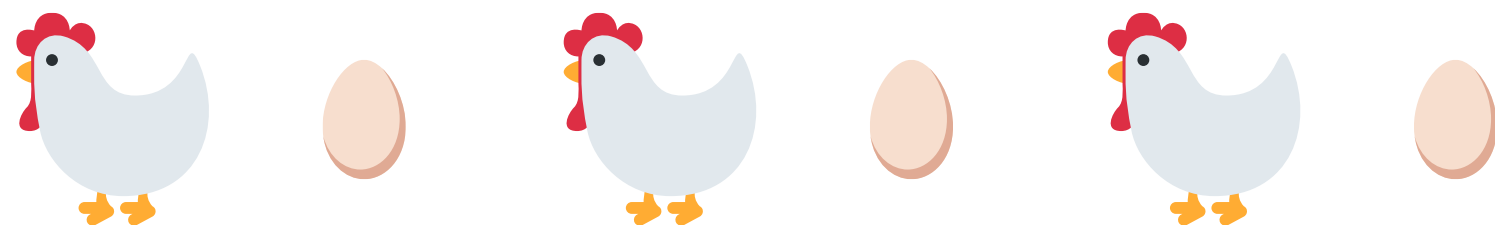
planned lumber



planed lumber
dressed lumber



wide piece lumber 4-5 in thick

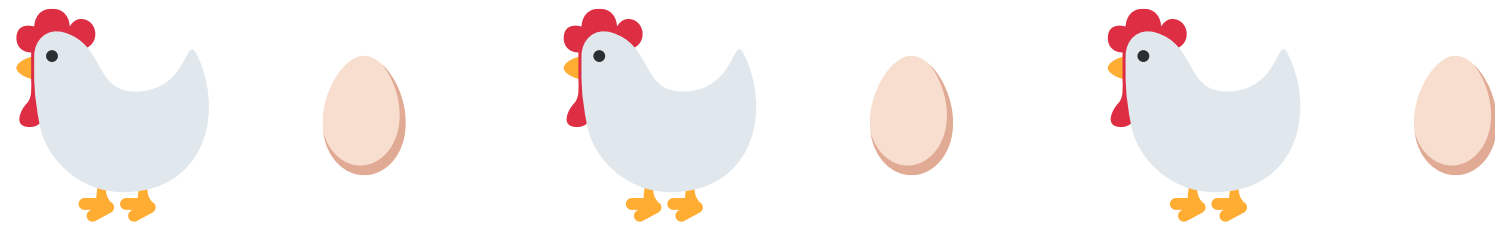


wide piece lumber 4-5 in thick

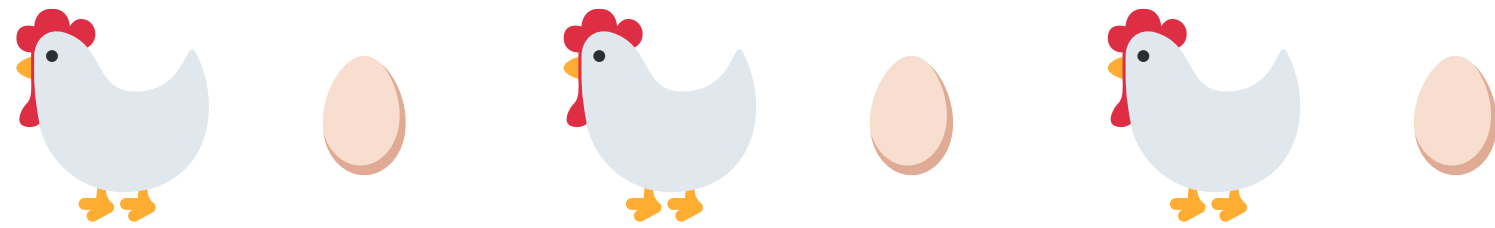
plank

tularan natin
kapatid



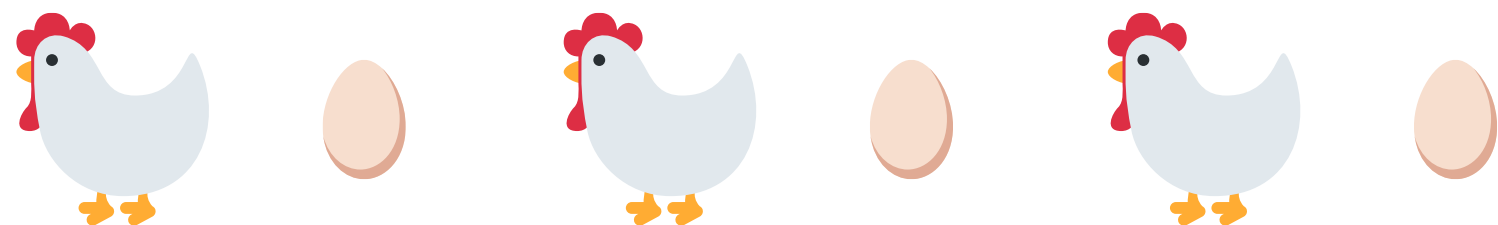


low density board from sawdust, chips,
and shavings



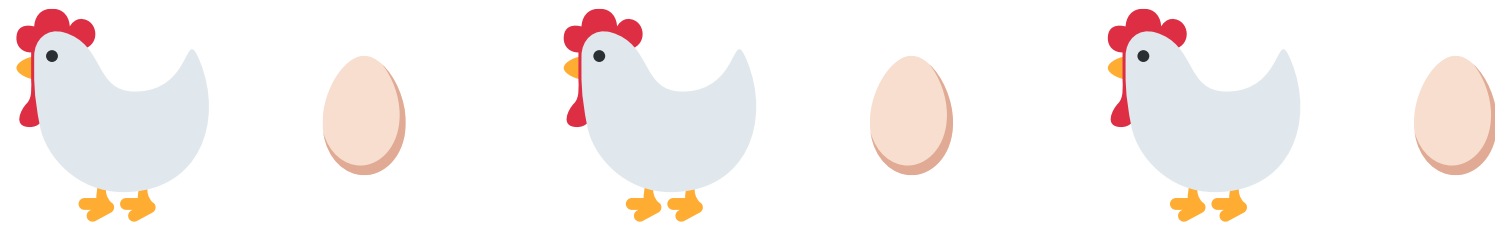
low density board from sawdust, chips,
and shavings

particleboard



identify





identify

**tounge and groove
(T&G)**



nainternalize
na?



A man and a woman are shown in a dimly lit room. The man, wearing a blue shirt, is in the foreground, looking down with a somber expression. The woman is partially visible behind him, looking towards the camera with a slight smile. The background is dark and out of focus, suggesting an indoor setting with some light sources.

30 Minutes Break

Jo, ano? Ayaw mo na?
Pagod ka na?

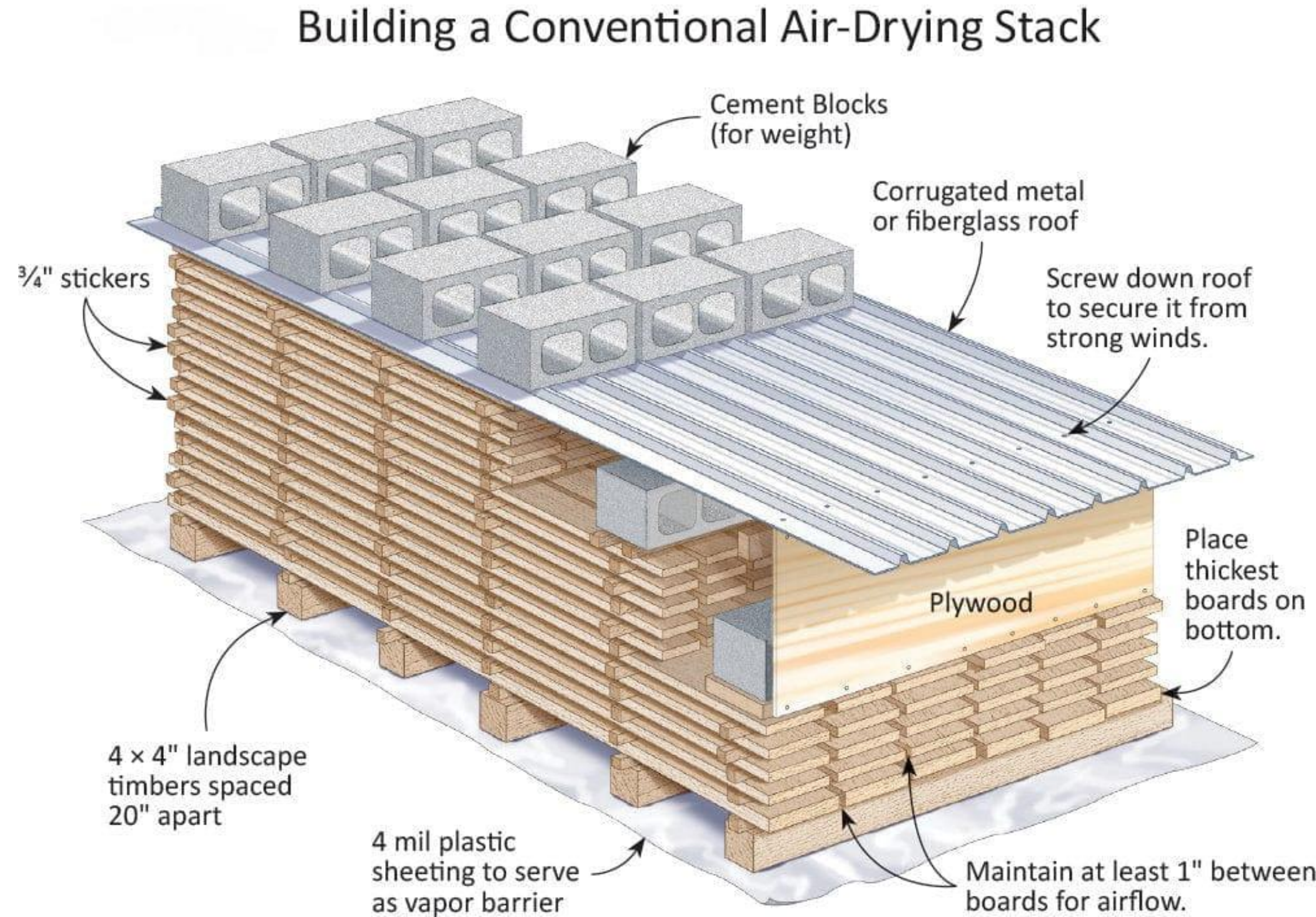
Lumber Seasoning

- removing excess moisture from freshly cut wood to make it suitable for use in construction, furniture, or other applications
- prevents wood from shrinking and warping



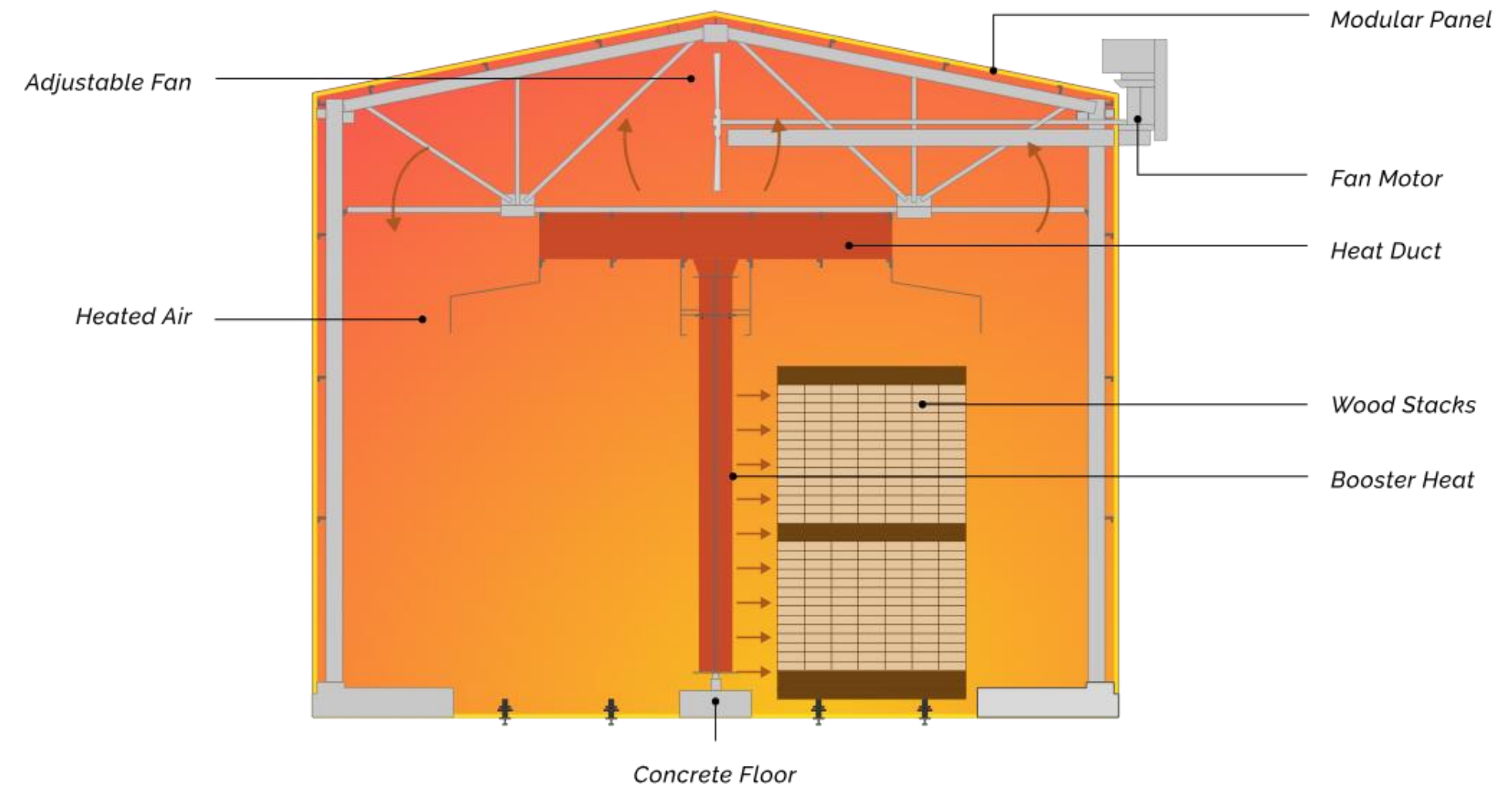
Air Drying

- lumber is stacked in open air with spacers (stickers) to allow airflow
- Simple and cheap, but slow (can take months to years)



Kiln Drying

- Lumber is dried in a kiln with controlled temperature, humidity, and air circulation
- Much faster (days to weeks) and produces uniform results





bow crook kink cup twist

Clean Development Mechanism

- allows developed countries to fund greenhouse gas emissions-reduction projects in developing countries
- developed countries earn a Certified Emission Reduction (CER) equivalent to one metric ton of CO₂e which developed countries can use toward meeting their emission reduction obligations



Kyoto Protocol

- the first legally binding agreement among countries to reduce greenhouse gas (GHG) emissions and combat global warming
- creates the CDM

Kyoto Protocol

first binding agreement to tackle global warming

Doha Amendment

2nd update or extension of the Kyoto protocol
2nd Commitment Period to reduce GHG emission

Paris Agreement

very latest update to mitigate global warming
the goal is to limit 1.5oC below levels of 1990

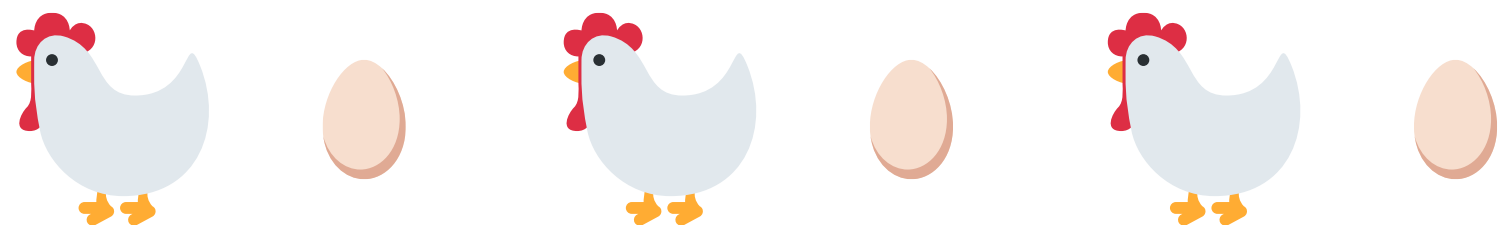
GWP

- global warming potential
- a measure of how much heat a greenhouse gas traps in the atmosphere compared to carbon dioxide (CO₂) over a specific period of time (usually 100 years)

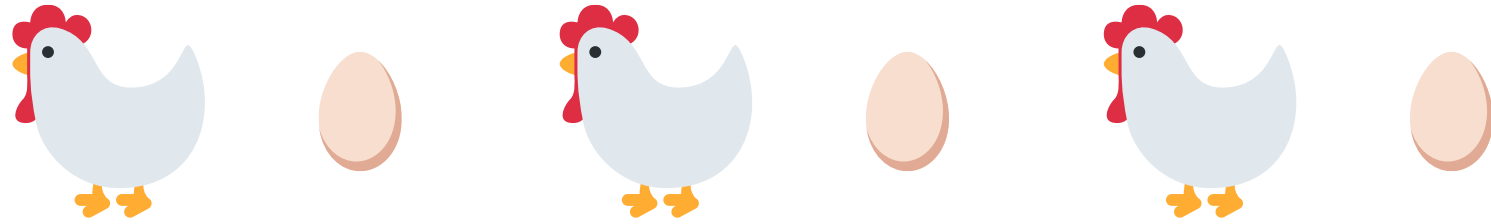
Industrial designation or common name	Chemical formula	GWP values for 100-year time horizon		
		Second Assessment Report (SAR)	Fourth Assessment Report (AR4)	Fifth Assessment Report (AR5)
Carbon dioxide	CO ₂	1	1	1
Methane	CH ₄	21	25	28
Nitrous oxide	N ₂ O	310	298	265

A man with dark hair, wearing a white button-down shirt, is pointing his right index finger upwards. He has a serious expression and is looking slightly to the right of the camera. The background is plain white.

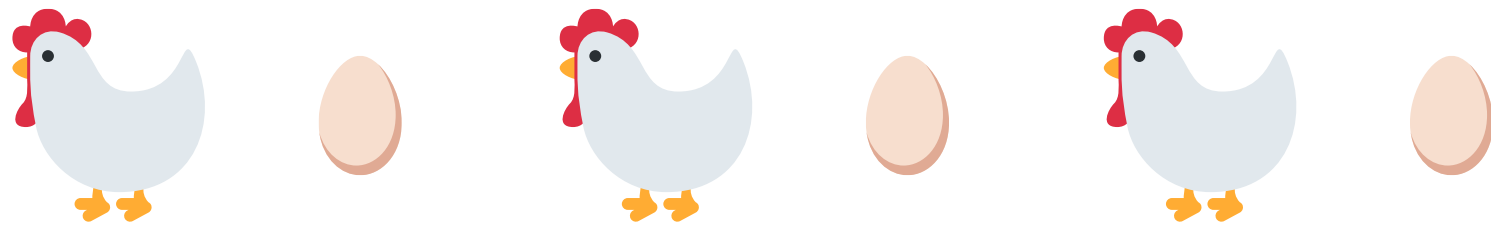
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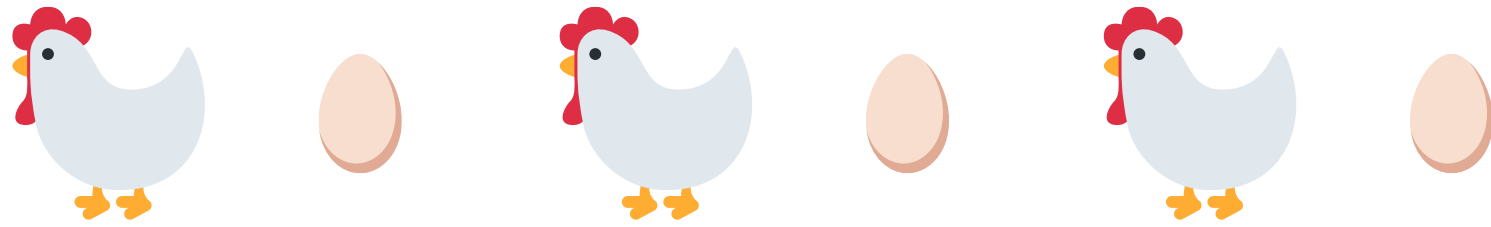
drying of lumber



drying of lumber
lumber seasoning

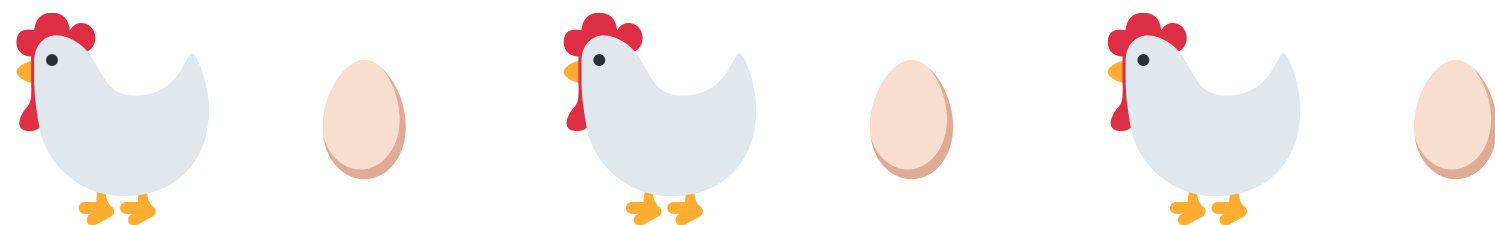


lumber is dried in open air

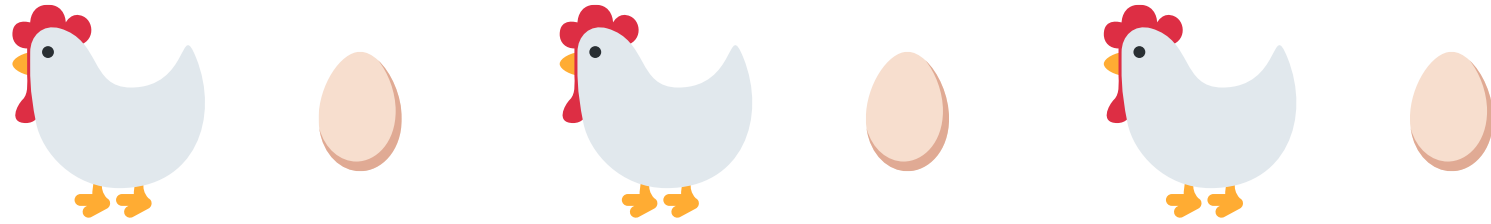


lumber is seasoned in open air

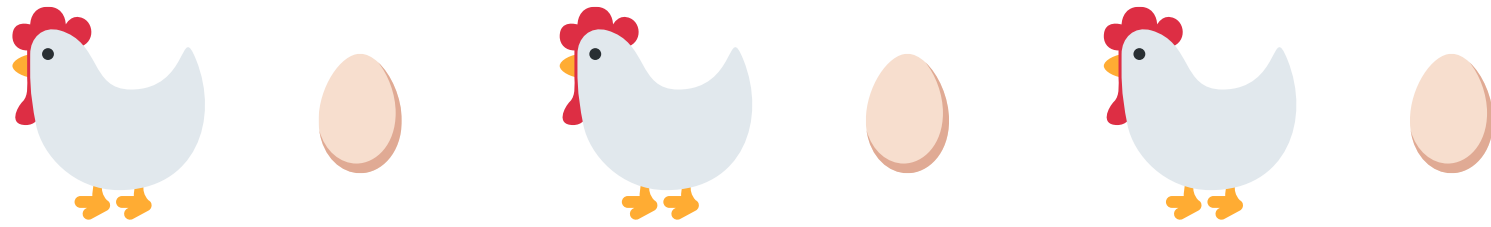
air drying



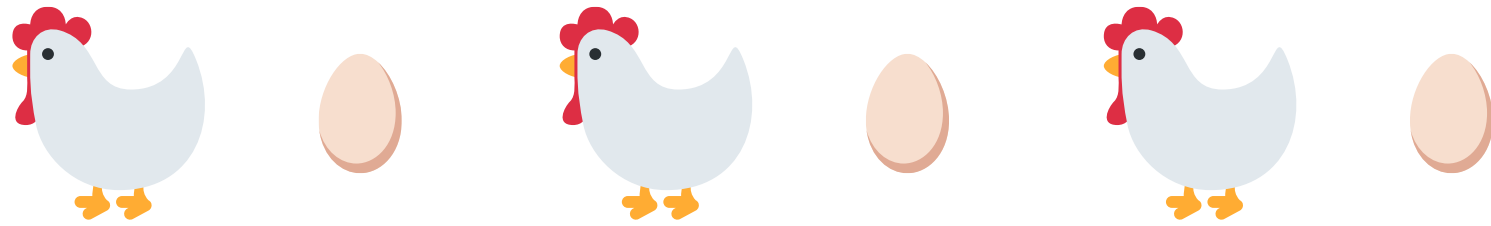
lumber is dried in a controlled
environment



lumber is dried in a controlled
environment
kiln drying

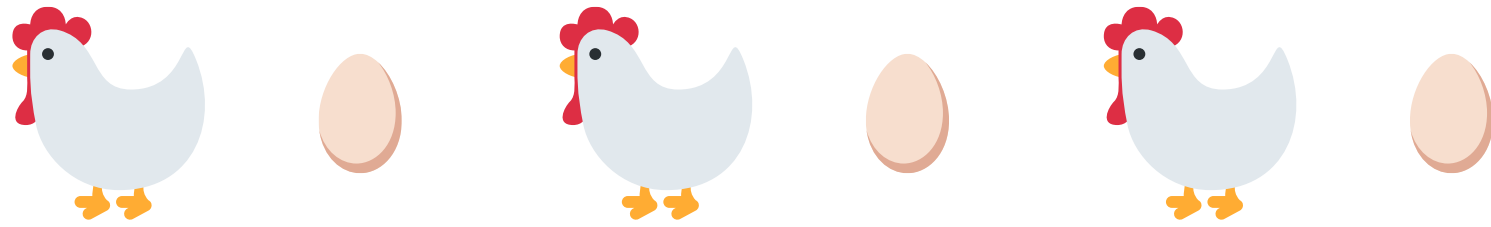


uneven drying of the wood results to

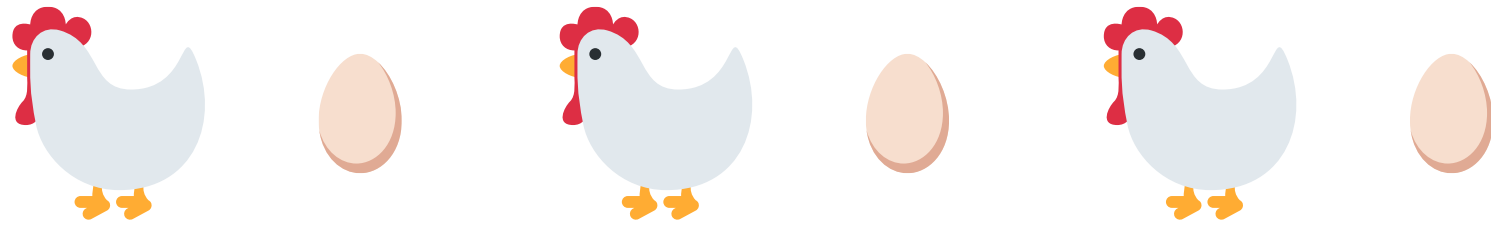


uneven drying of the wood results to

warping

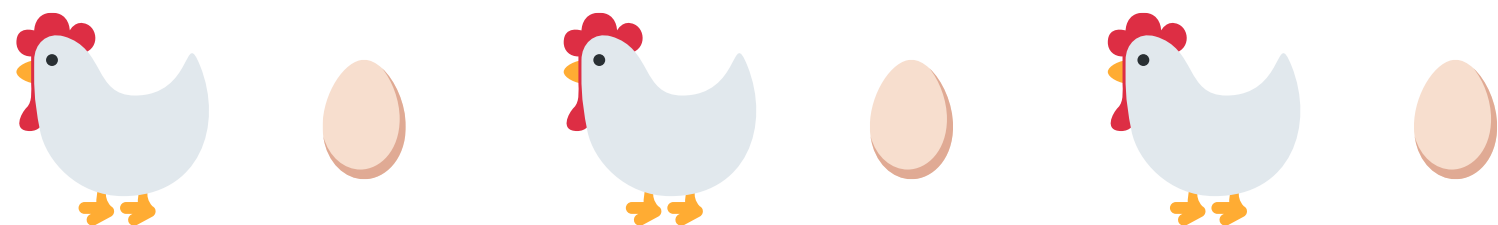


loss of water contained in the cell wall



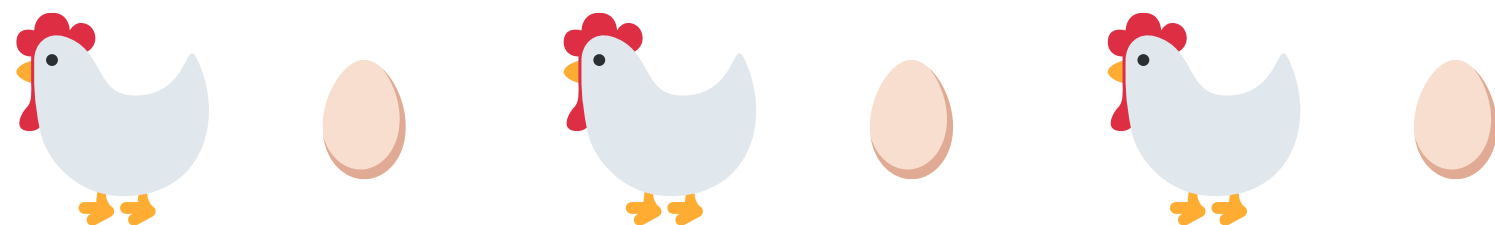
loss of water contained in the cell wall

shrinkage



identify

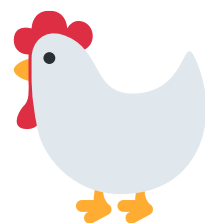
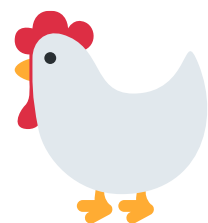
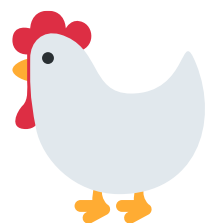




identify

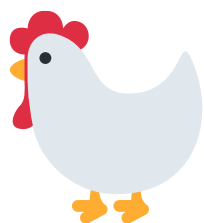
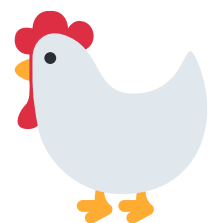
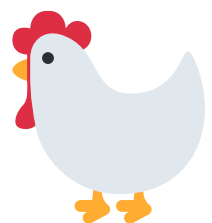
crook





identify

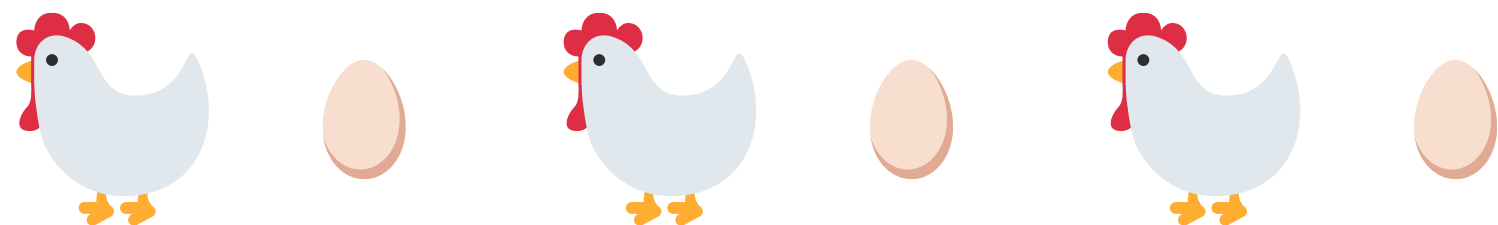




identify

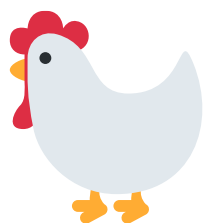
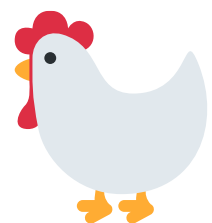
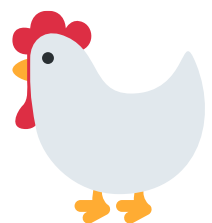
cup





identify

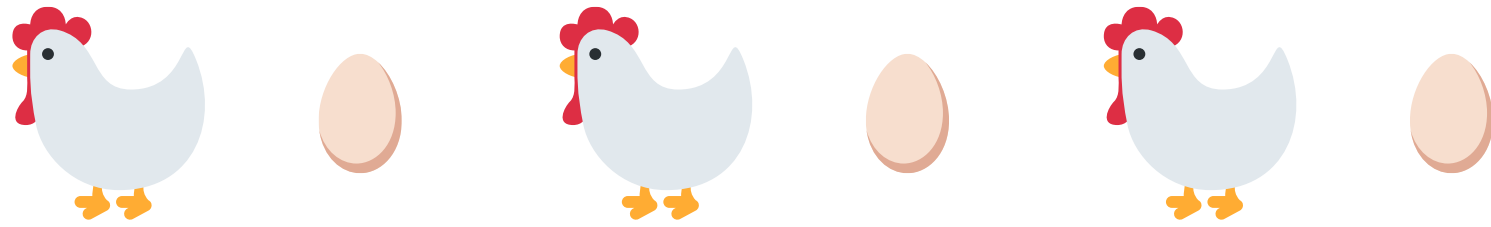




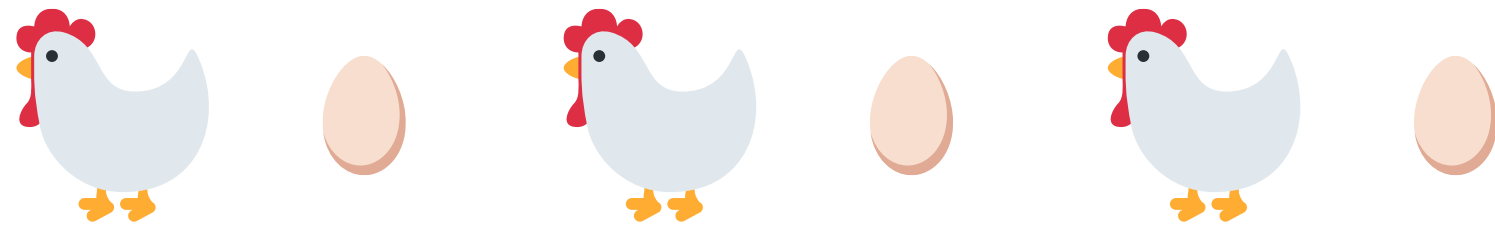
identify

bow



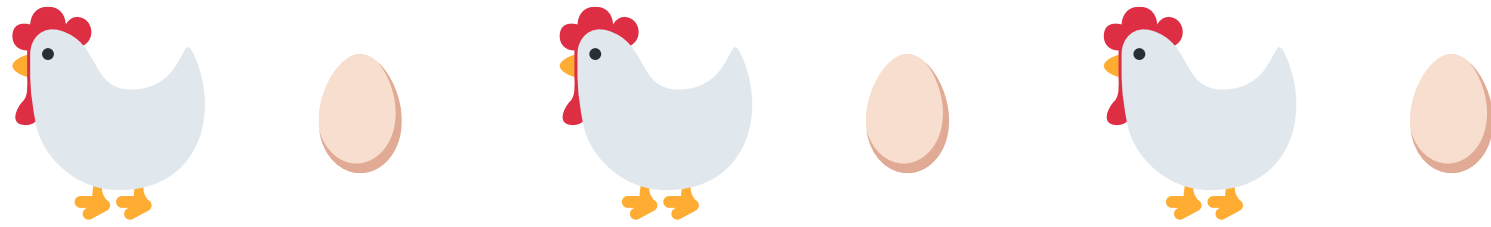


measure of how much a gas traps heat
compared to carbon dioxide

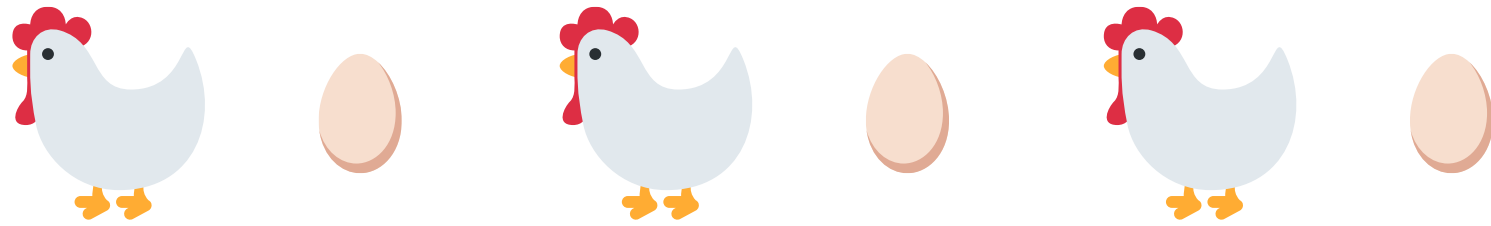


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GWP

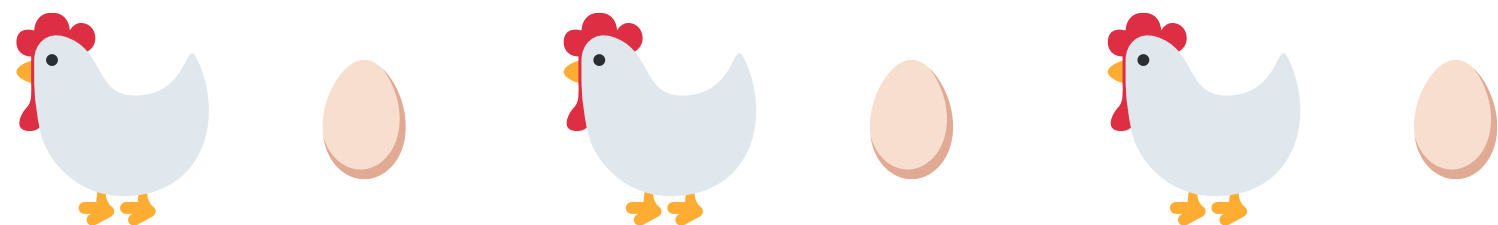


what is GWP

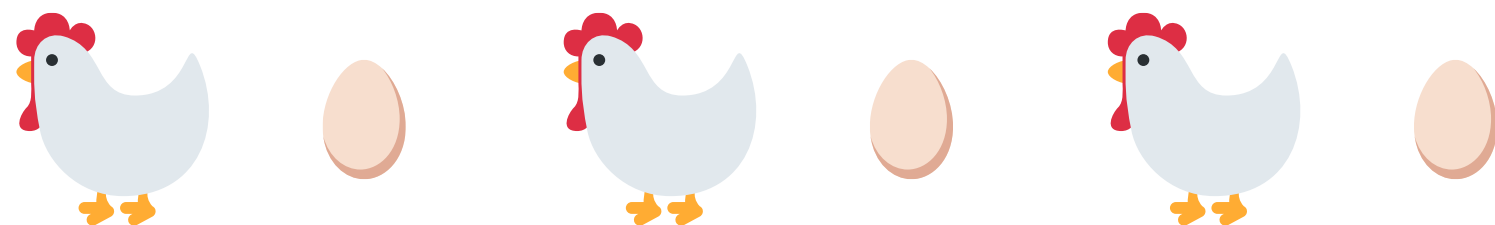


what is GWP

global warming potential



the agreement that extends Kyoto
Protocol



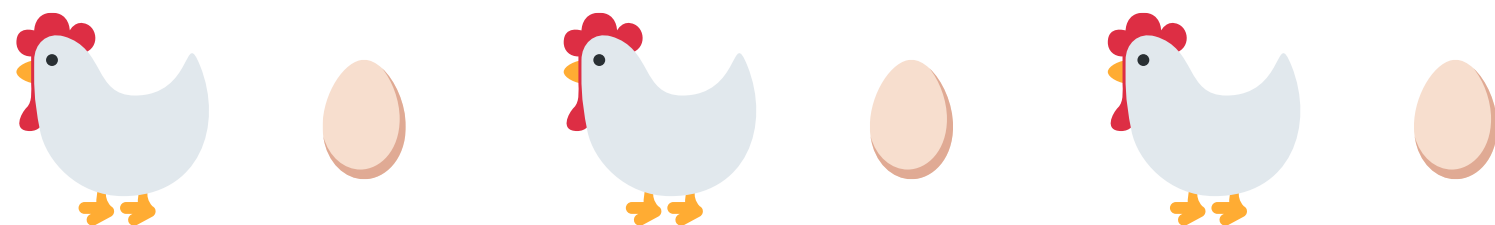
the agreement that extends Kyoto
Protocol

Doha Amendment

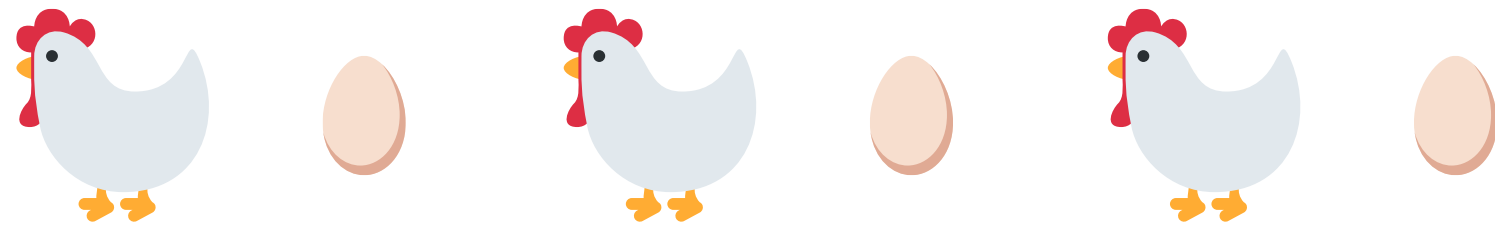


@6xmta
freshman college:



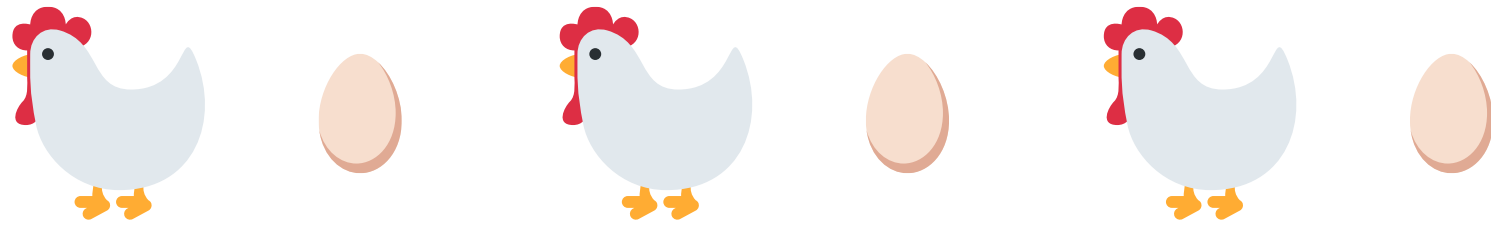


The convention that wants to limit
global warming to at least 1.5C below
1990 levels

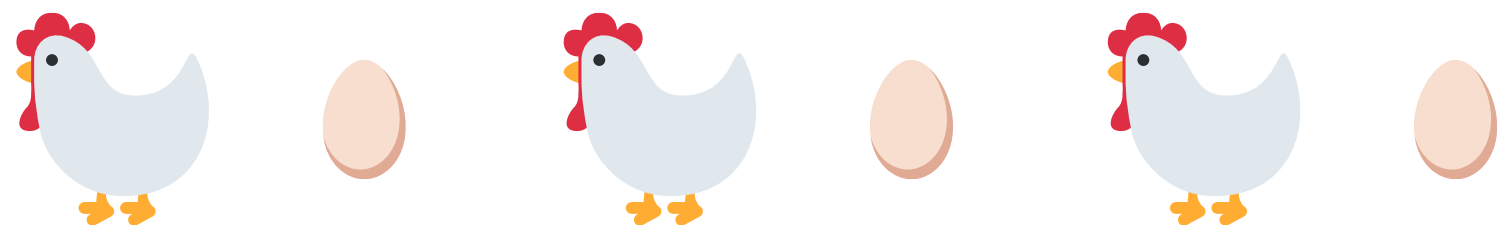


The convention that wants to limit
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Paris Agreement

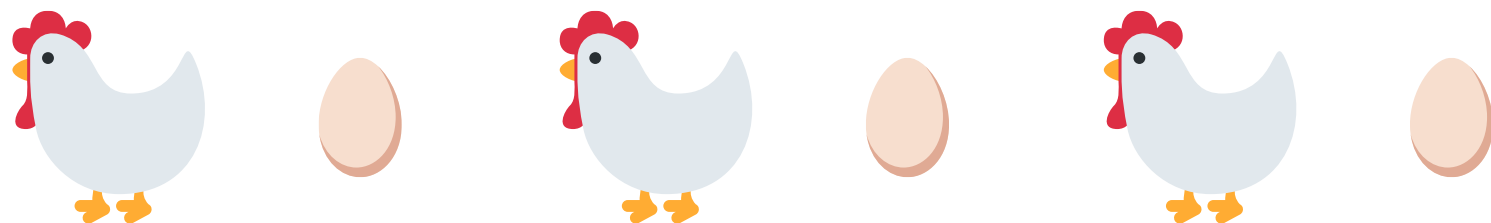


mechanism created by Kyoto Protocol

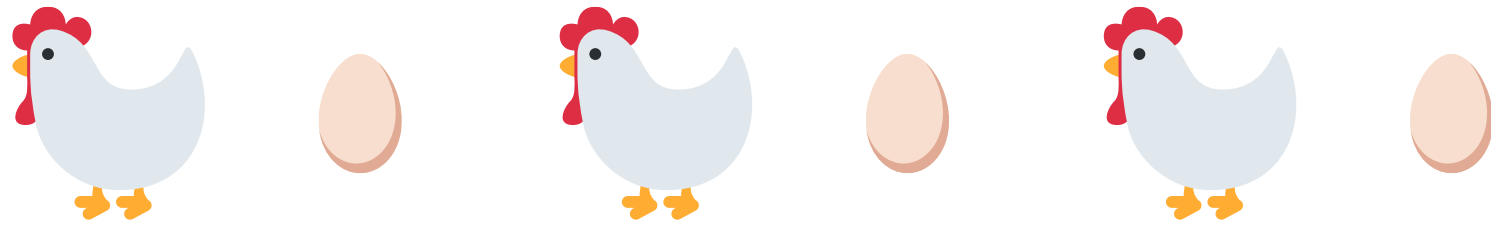


mechanism created by Kyoto Protocol

CDM



what is CDM



mechanism created by Kyoto Protocol

clean development mechanism

A rice mill operates 5 hours per day, 7 days per week, and processes 500 kg of rice per hour. If the emission factor for particulate matter is 5 lb/ton and no pollution control facility is used, what is the estimated annual particulate matter emissions?

Remember this equation:

$$E = A \times EF \left(1 - \frac{ER}{100} \right)$$

Where E is the emission, *ton/yr*

A is the operation or activity

EF is the emission factor

ER is the emission reduction practices

Next, just do unit analysis and plugging in the values in the equation.

Our equation is simplified because no ER is done in the problem.

$$E = A \times EF$$

$$E = 500 \frac{\text{kg}}{\text{hr}} \times 5 \frac{\text{hr}}{\text{day}} \times 7 \frac{\text{days}}{\text{week}} \times 52 \frac{\text{weeks}}{\text{yr}} \times 5 \frac{\text{lb}}{\text{ton}} \times \frac{1 \text{ ton}}{2200 \text{ lb}} \times \frac{1 \text{ ton}}{1000 \text{ kg}} = \mathbf{2.07 \text{ tons/yr}}$$

A poultry farm operates with 30,000 birds per grow cycle, with 5 cycles per year, applies manure to land without any treatment. Using N₂O emission factor of 0.6 kg/head/year, adjusted for grow time (38 days), what is the estimated nitrous oxide emission?

Remember this equation:

$$E = A \times EF \left(1 - \frac{ER}{100} \right)$$

Where E is the emission, *ton/yr*

A is the operation or activity

EF is the emission factor

ER is the emission reduction practices

$$E = A \times EF$$

$$E = 30,000 \text{ birds} \times 5 \frac{\text{cycles}}{\text{yr}} \times 38 \frac{\text{days}}{\text{cycle}} \times \frac{1 \text{ yr}}{365 \text{ days}} \times 0.6 \frac{\text{kg}}{\text{bird} \cdot \text{yr}} \times \frac{1 \text{ ton}}{1000 \text{ kg}} = \mathbf{9.37 \text{ tons N}_2\mathbf{O/yr}}$$

From the previous problem, what is the CO₂ equivalent using the GWP of nitrous oxide = 298?

$$E = A \times EF$$

$$E = 30,000 \text{ birds} \times 5 \frac{\text{cycles}}{\text{yr}} \times 38 \frac{\text{days}}{\text{cycle}} \times \frac{1 \text{ yr}}{365 \text{ days}} \times 0.6 \frac{\text{kg}}{\text{bird} \cdot \text{yr}} \times \frac{1 \text{ ton}}{1000 \text{ kg}} = \mathbf{9.37 \text{ tons N}_2\mathbf{O/yr}}$$

Remember that the GWP of N₂O is 298. This means that 1 molecule of N₂O equals 298 times CO₂.

$$E = 9.37 \frac{\text{tons N}_2\text{O}}{\text{yr}} \times \frac{298 \text{ tons CO}_2\text{e}}{1 \text{ ton N}_2\text{O}} = \mathbf{2,792.26 \frac{\text{tons CO}_2\text{e}}{\text{yr}}}$$

Problems Related to Rural Electrification and AB Structures



A box with a weight of 200 kg is lifted by a crane and is accelerating at 2 m/s^2 . What amount of force is being utilized by the crane to lift the box?

- A. 400 N
- B. 1960 N
- C. 800 N
- D. 2360 N



Solution

$$\Sigma F = ma$$

$$F_{lift} - W_{box} = m_{box}a_{box}$$

$$F_{lift} = m_{box}a_{box} + W_{box} = m_{box}(a_{box} + g)$$

$$F_{lift} = 200 \text{ kg} (2 \text{ m/s}^2 + 9.81 \text{ m/s}^2)$$

$$F_{lift} = \mathbf{2,362 \text{ N}}$$



A warehouse will install pendant dome incandescent lamps at a mounting height ratio of 1.50. The lamp will be mounted on a grid measuring 5 m by 5 m. What is the minimum height of the lamps?

- A. 3 m
- B. 3.30 m
- C. 7.5 m
- D. 7 m



Solution

$$\text{Mounting Ratio} = \frac{\text{Spacing}}{\text{mounting height}}$$

$$\begin{aligned}\text{mounting height} &= \frac{\text{Spacing}}{\text{Mounting Ratio}} \\ &= \frac{5 \text{ m}}{1.50} = \mathbf{3.33 \text{ m}}\end{aligned}$$



What is the total general lighting load for a two-storey residence having a dimension of 6 m by 9 m?

- A.1728 VA
- B.3456 VA
- C.1620 VA
- D.3240 VA



Solution

Remember that PEC specify a general lighting load to have 32 VA/m².

$$\begin{aligned} GLL &= 32 \frac{\text{VA}}{\text{m}^2} \times (6 \text{ m} \times 9 \text{ m}) \times 2 \\ &= \mathbf{3,456 \text{ VA}} \end{aligned}$$



A 40 W fluorescent lamp produces 3,000 lumens of light in a room with dimensions of 3 m by 5 m. What is the illumination of the floor?

- A. 200 lux
- B. 40 fc
- C. 200 fc
- D. 50 lux



Solution

$$\begin{aligned} \text{Illumination} &= \frac{\text{lumen}}{\text{area}} \\ &= \frac{3000 \text{ lumen}}{3 \text{ m} \times 5 \text{ m}} \\ &= \mathbf{200 \text{ lux}} \end{aligned}$$



Compute the brightness of a fixture with a 1' by 4' plastic diffuser having a transmittance of 0.6 and illuminated by 2 pcs 3200 lumen lamps, assuming 100% use of light flux.

- A. 1033 millilambert
- B. 1600 millilambert
- C. 800 millilambert
- D. 960 millilambert



Solution

$$\begin{aligned} \text{Luminance (Brightness)} &= \text{illumination} \times \text{factor} \\ &= \frac{3200 \text{ lumen}}{1 \text{ ft} \times 4 \text{ ft}} \times 0.60 \times 2 \\ &= 960 \text{ footlambert} \end{aligned}$$

Remember that 1 footlambert = 1.076 millilambert

$$\begin{aligned} \text{Luminance} &= 960 \text{ footlambert} \times \frac{1.076 \text{ millilambert}}{1 \text{ footlambert}} \\ &= \mathbf{1,032.96 \text{ millilambert}} \end{aligned}$$



A 40 W fluorescent lamp 120 cm long produces 3200 lumens of light in a having a general dimension of 15 ft by 20 ft. Find the illumination.

- A. 10.67 lumen
- B. 10.67 footlambert
- C. 10.67 lux
- D. 10.67 footcandle



A 40 W fluorescent lamp 120 cm long produces 3200 lumens of light in a having a general dimension of 15 ft by 20 ft. Find the illumination.

A.10.67 lumen

B.10.67 footlambert

C.10.67 lux

D.10.67 footcandle



From the previous problem, what is the luminance if the reflectance factor of the walls are 40%?

- A. 4.27 lumen
- B. 4.27 footlambert
- C. 4.27 lux
- D. 4.27 footcandle



From the previous problem, what is the luminance if the reflectance factor of the walls are 40%?

A.4.27 lumen

B.4.27 footlambert

C.4.27 lux

D.4.27 footcandle



A 120 V generator delivers 3 kW to an electric furnace. The current supplied by the generator is about:

- A. 2.5 A
- B. 250 mA
- C. 2500 mA
- D. 25000 mA



A step-down transformer has a primary voltage of 120 V and a secondary voltage of 24 V. If a current of 5 A flows on the secondary, what is the current flowing on the primary side?

- A. 2 A
- B. 24 A
- C. 25 A
- D. 1 A



If a transformer has a ratio of one primary turn to 10 secondary turns, what is the secondary current for a primary current of 40 A assuming that there is no power loss?

- A. 2 A
- B. 4 A
- C. 6 A
- D. 8 A



What is the maximum fixture spacing of a room that has a ceiling height of 3 m and is to be lighted with directly concentrating fluorescent lamps (0.40)?

- A. 1.1 m
- B. 1.3 m
- C. 3.84 ft
- D. 3.94 ft



Determine the number of poles on the rotor of a single-phase generator if the frequency of 60 Hz is generated at a rotor speed of 1800 rpm.

- A. 2 poles
- B. 4 poles
- C. 6 poles
- D. 8 poles



What would be the speed of a 120 V
6-pole motor on a 60 Hz source?

- A. 1800 rpm
- B. 1200 rpm
- C. 900 rpm
- D. 2400 rpm



What is the equivalent area of No. 8 conductor wire in square inches?

- A. 0.1 square in
- B. 0.013 square in
- C. 0.01 square in
- D. 1.0 square in



What is the equivalent size in square mm of a cable 250 MCM?

- A. 127 mm²
- B. 126 mm²
- C. 125 mm²
- D. 124 mm²



A mercury lamp having a hot resistance of 50 ohms, is connected to a socket with 240 V current supply. How much current flows through the lamp?

- A. 5 A
- B. 4.8 A
- C. 4 A
- D. 5.8 A



A circuit has a resistance of 20 ohms and the current flows at 12 amperes. Determine the voltage.

- A. 240 V
- B. 120 V
- C. 230 V
- D. 150 V



An electric motor has a trademark label of 2 hp, 240 V, 15 A. Calculate the motor power factor. Assume 85% motor efficiency.

- A. 0.75
- B. 0.50
- C. 0.80
- D. 0.66



What is the power factor for a purely inductive circuit?

- A.1
- B.0
- C.<1
- D.>1



An electric motor draws 2,304 W and 12 A when operating on a 240 V, 60 Hz source. What is the power factor of the motor?

- A. 0.75
- B. 0.80
- C. 0.50
- D. 0.66



A hollow steel tube with an inside diameter of 100 mm must at tensile load of 400 kN. Determine the outside diameter of the tube if the stress is limited to 120 MPa.

A.110 mm

B.119 mm

C.200 mm

D.210 mm



The tensile stress that will be developed if a force of 10 N is applied to one end of a 10 inches diameter circular rod.

A. 0.10 kPa

B. 0.05 kPa

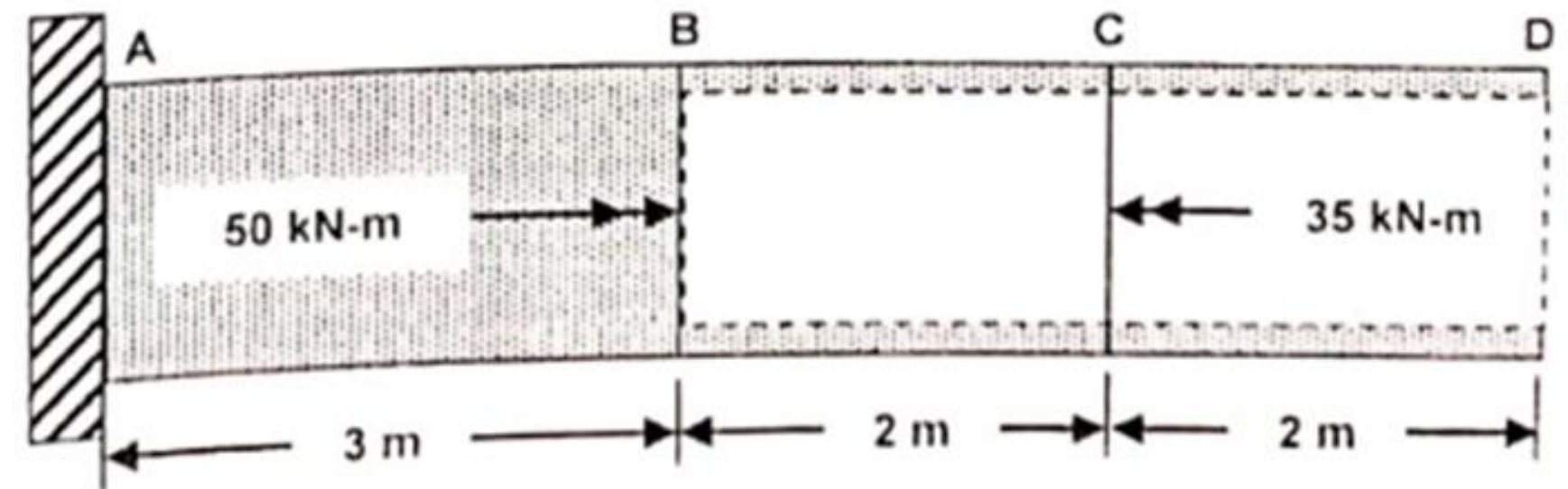
C. 0.20 kPa

0.15 kPa



The 200-mm diameter steel shaft ($G=80$ GPa) has a 150-mm diameter bronze core ($G=40$ GPa) inserted in 4 m of the right end and securely bonded to the steel. When torque of 50 and 35 kN-m are applied as shown, determine the maximum shearing stress in bronze.

- A. 8.59 MPa
- B. 9.93 MPa
- C. 25.77 MPa
- D. 40.16 MPa





Solution

At points B and C, we have a composite shafting. The torque on those points is split.

Remember this:

$$\tau = \frac{T \cdot r}{J} \qquad \theta = \frac{TL}{JG}$$

It is assumed that the angle of twist is the same for both material and they have the same length. Then,

$$\left(\frac{\theta}{L}\right)_{st} = \left(\frac{\theta}{L}\right)_{br}$$

$$\frac{T_{st}}{J_{st}G_{st}} = \frac{T_{br}}{J_{br}G_{br}}$$

$$\frac{T_{st}}{T_{br}} = \frac{J_{st}G_{st}}{J_{br}G_{br}}$$



Solution

Remember also at that point the total torque of the two components equal the torque at that point.

$$T = T_{st} + T_{br}$$

Since the bronze is our concern, let us equate the equation so that only the bronze variable remains.

$$T_{st} = \frac{J_{st} G_{st}}{J_{br} G_{br}} T_{br} = \frac{\frac{\pi}{32} (200^4 - 150^4) \text{ mm}^4 \times 80 \text{ GPa}}{\frac{\pi}{32} (150^4) \text{ mm}^4 \times 40 \text{ GPa}} T_{br}$$

$$T_{st} = 4.32 T_{br}$$

Substitute the value of T_{st}

$$T = 4.32 T_{br} + T_{br} = 5.32 T_{br}$$

$$T_{br} = \frac{T}{5.32} = \frac{35}{5.32} \text{ kN} \cdot \text{m} = 6.578 \text{ kN} \cdot \text{m}$$



Solution

Finding now the max shearing stress of bronze

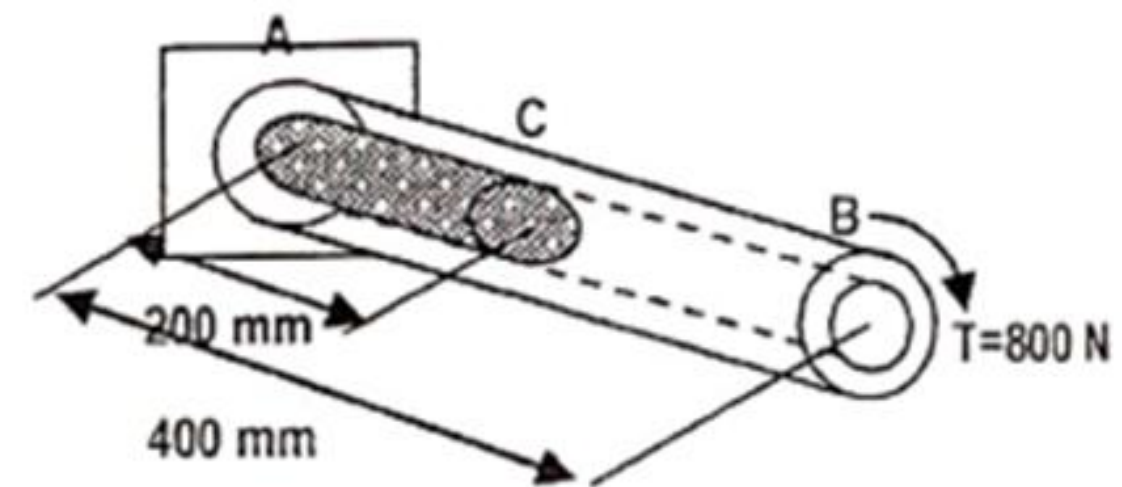
$$\begin{aligned}(\tau_{max})_{br} &= \frac{T \cdot r}{J} = \frac{6.58(1000)(1000) \text{ N} \cdot \text{mm} \times \left(\frac{150}{2}\right) \text{ mm}}{\frac{\pi}{32} (150)^4 \text{ mm}^4} \\ &= \mathbf{9.93 \text{ MPa}}\end{aligned}$$



The tube is made of 2014-T6 aluminum and has an inner diameter of 30 mm and an outer diameter of 55 mm. A torque of $T = 800 \text{ N-m}$ is applied to its end B. If its filled with an A-36 steel plug AC. Consider the values: $G_{\text{A-36 steel}} = 75 \text{ GPa}$, and $G_{\text{2014-T6 aluminum}} = 27 \text{ GPa}$.

Determine the maximum shear stress in the steel.

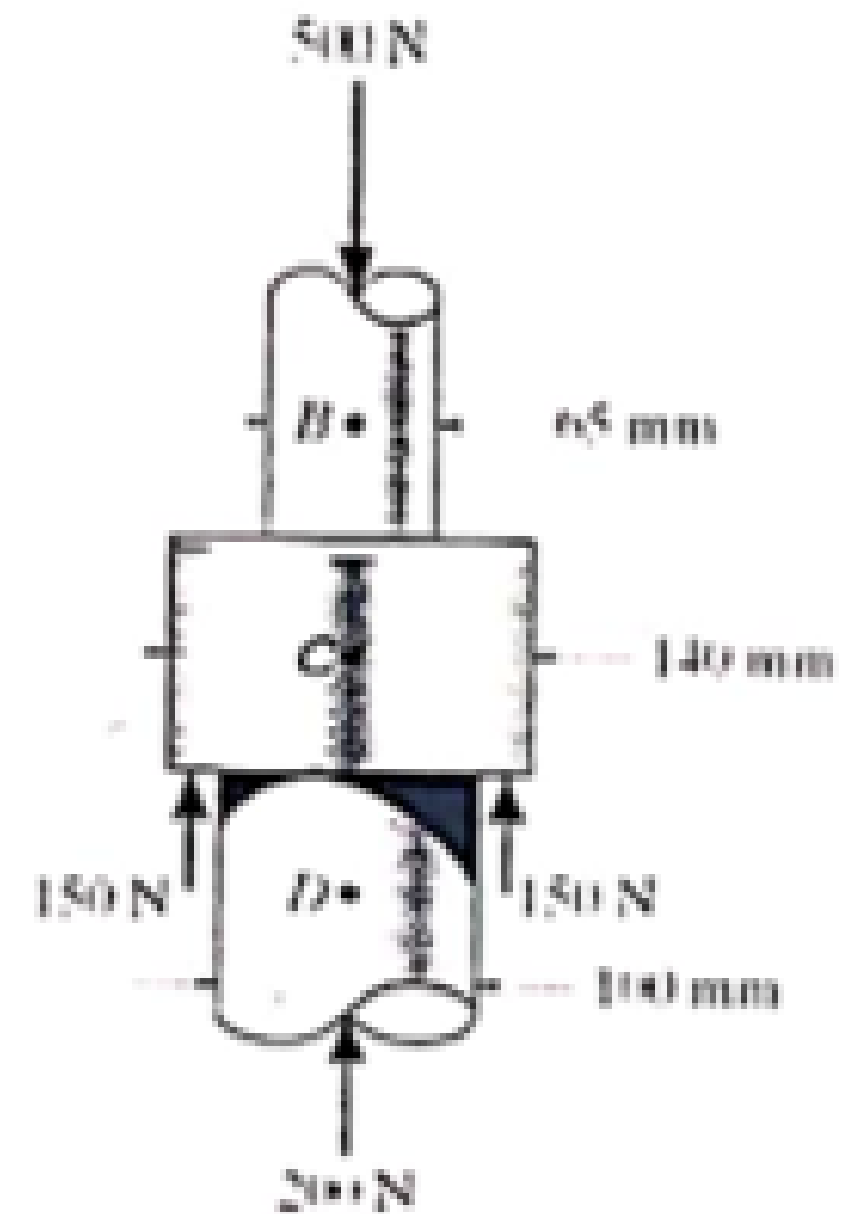
- A. 21.16 MPa
- B. 26.87 MPa
- C. 169.96 MPa
- D. 32.06 MPa





The thrust bearing is subjected to the loads shown.
Determine the average normal stress developed on the cross-section through point C.

- A. 25.5 kPa
- B. 32.5 kPa
- C. 150.7 kPa
- D. 63.7 kPa





Solution

$$\sigma = \frac{P}{A}$$

$$\sigma = \frac{500 \text{ N}}{\frac{\pi}{4} (140)^2 \text{ mm}^2} = 0.0325 \text{ MPa}$$

$$\sigma = 0.0325 \text{ MPa} \times \frac{1000 \text{ kPa}}{1 \text{ MPa}} = \mathbf{32.5 \text{ kPa}}$$



A wooden rectangular frame (1.5 m x 3 m) is to be fabricated using 2" x 3" coco lumber. Compute for the total bdft required for this frame.

- A.15
- B.10
- C.8
- D.6



Solution

First get the total length of the wooden frame by getting its perimeter.

$$P = 2(l + w) = 2(1.5 \text{ m} + 3 \text{ m}) = 9 \text{ m}$$

Convert the length to ft.

$$9 \text{ m} \times \frac{1 \text{ ft}}{0.3048 \text{ m}} = \mathbf{29.5 \text{ ft}}$$

Solve for bdft:

$$bdft = \frac{2 \text{ in} \times 3.5 \text{ in} \times 29.5 \text{ ft}}{12} = \mathbf{14.75 \text{ bdft}}$$



A wooden rectangular frame (2 m x 3.5 m) is to be fabricated using 2" x 3" coco lumber. Compute for the total bdft required for this frame.

- A.25
- B.19
- C.10
- D.7



Solution

First get the total length of the wooden frame by getting its perimeter.

$$P = 2(l + w) = 2(2 \text{ m} + 3.5 \text{ m}) = 11 \text{ m}$$

Convert the length to ft.

$$11 \text{ m} \times \frac{1 \text{ ft}}{0.3048 \text{ m}} = \mathbf{36.09 \text{ ft}}$$

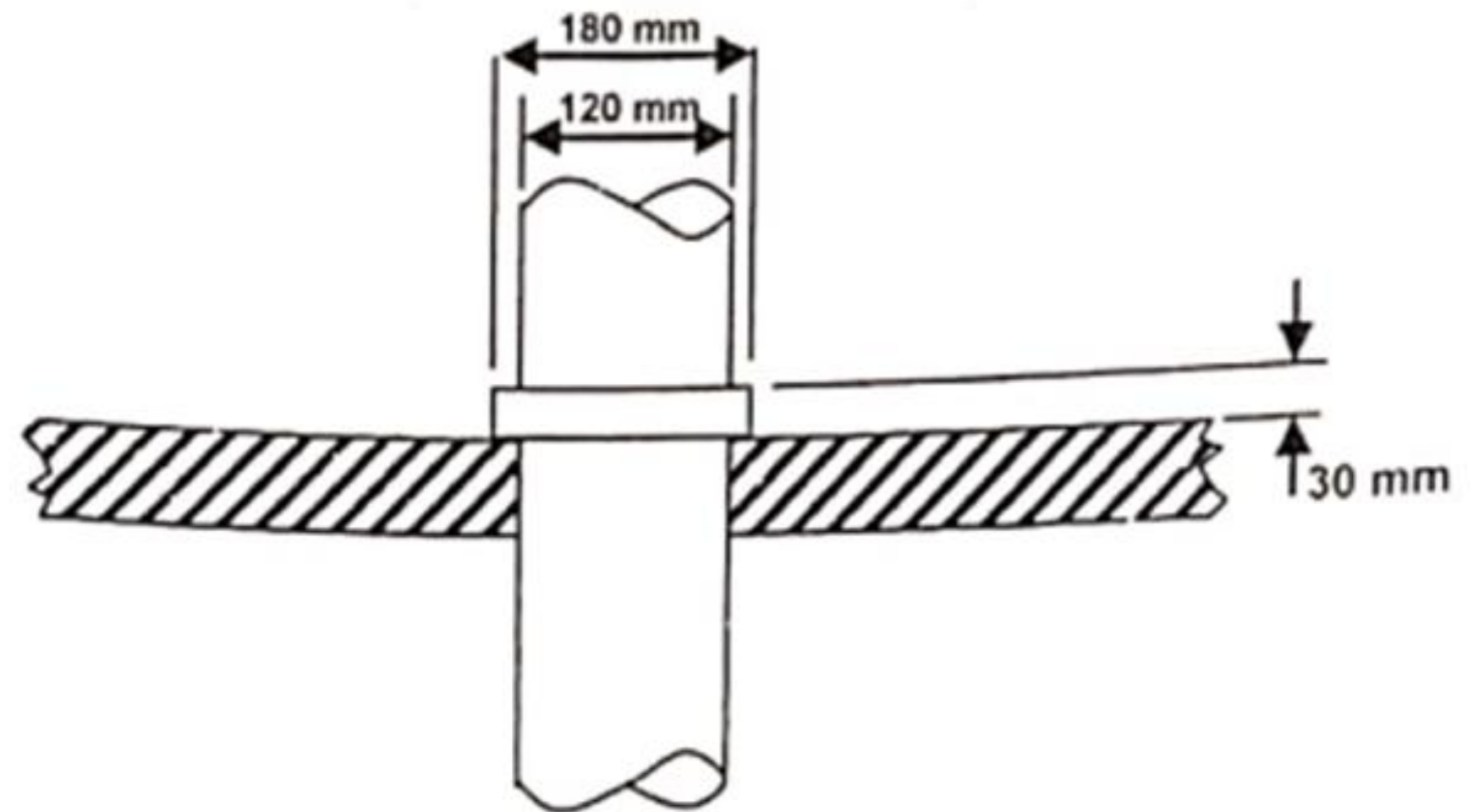
Solve for bdft:

$$bdft = \frac{2 \text{ in} \times 3 \text{ in} \times 36.09 \text{ ft}}{12} = \mathbf{18 \text{ bdft}}$$



A vertical shaft is supported by a thrust collar and bearing plate. Determine the actual bearing stress between the collar and the plate if an axial load of 1000 kN is applied to the shaft.

- A. 39.30 MPa
- B. 58.95 MPa
- C. 62.35 MPa
- D. 70.73 MPa





Solution

Remember the formula for bearing stress,

$$\sigma = \frac{P}{A} = \frac{1000(1000) \text{ N}}{\frac{\pi}{4} (180)^2 \text{ mm}^2} = \mathbf{39.30 \text{ MPa}}$$



The beam has a rectangular cross-section and is made of wood having an allowable shear stress of 1.6 ksi. If it is subjected to a shear of 4 kips, determine the smallest dimension of a if its bottom and $1.5a$ if its sides.

- A. 1.08 in
- B. 1.58 in
- C. 2.08 in
- D. 2.58 in



Solution

Remember this equation for a rectangular cross section.

$$\tau = \frac{3V}{2A}$$

$$A = \frac{3V}{2\tau}$$

$$a(1.5a) = \frac{3(4000) \text{ lb}}{2(1.6)(1000) \frac{\text{lb}}{\text{in}^2}}$$

$$1.5a^2 = 3.75 \text{ in}^2$$

$$a = \sqrt{\frac{3.75 \text{ in}^2}{1.5}} = \mathbf{1.58 \text{ in}}$$



A vertical shaft is supported by thrust collar and bearing plate. Determine the maximum axial load that can be applied to the shaft if the average punching shear stress in the collar and the average bearing stress between the collar and the plate are limited to 50 and 90 MPa, respectively. The diameter of the bearing plate is 180 mm, diameter of the vertical shaft is 120 mm, and thickness of bearing plate is 30 mm.

- A. 565.49 kN
- B. 848.23 kN
- C. 1130.97 kN
- D. 1272.35 kN



Solution

Solved separately the axial force for both stresses, and the one that is smaller in value is the maximum safe load.

$$\sigma = \frac{P}{A} \Rightarrow P = \sigma A = 90 \frac{\text{N}}{\text{mm}^2} \times \frac{\pi}{4} (180)^2 \text{mm}^2 \\ = 2,290,221 \text{ N or } 2,290.2 \text{ kN}$$

$$\tau = \frac{P}{A} \Rightarrow P = \tau A = 50 \frac{\text{N}}{\text{mm}^2} \times \pi (120 \text{ mm})(30 \text{ mm}) \\ = \mathbf{565,486.68 \text{ N or } 565.49 \text{ kN}}$$

Choose 565 kN as the maximum because more than this load the plate will fail.